

On the Global Compactness of Palais-Smale Sequences for p -Laplace Equations with Critical Nonlinearities in Smoothly Bounded Domains

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ABSTRACT. In this text, we give a mostly self-contained treatment of a global compactness result for p -Laplace equations with critical nonlinearities in $\mathbb{R}^N$. Namely, we will prove a representation theorem for (possibly sign changing) Palais-Smale sequences established by Meruci-Willem in 2010. Furthermore, we discuss the application of non-existence results to the classification of the resulting bubbles.

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1. Introduction

In these notes, we discuss some recent developments regarding the global compactness of Palais-Smale sequences for p -Laplace equations with critical nonlinearities. More precisely, we shall investigate the limiting behaviour of minimizing sequences to energy functionals for a large class of critical p -Laplace equations. In particular, we introduce a new argument due to Meruci-Willem [3] which simplifies the approach taken by Struwe in [5] and [6] for the $p = 2$ case. Put informally, the key step in this new argument will be to focus on the dual space of $\mathcal{D}^{1,p}(\mathbb{R}^N)$, a homogeneous Sobolev space, rather than the “averaged decay” of the given Palais-Smale sequence.

Global compactness results are now indispensable, and highly sought after tools, for the analysis of partial differential equations. Indeed, they can be used to establish the existence of non-trivial solutions to various problems (e.g. blow-up solutions to Schrödinger equations and solutions to Yamabe equations – see Palatucci-Pisante [4] for more details). However, despite their numerous applications, these compactness results are interesting in their own right. Fundamentally, they describe an asymptotic expansion of the Palais-Smale sequence in the energy space $\mathcal{D}^{1,p}(\mathbb{R}^N)$, which we shall introduce shortly.

Henceforth, $N \geq 3$ is an integer, $\mu > 0$ is any real number, and Ω is a bounded domain in $\mathbb{R}^N$ having smooth boundary $\partial\Omega$. Unless stated otherwise, we will assume that $1 < p < N$. We denote by p' the conjugate Hölder exponent of p , and by p^* the Sobolev conjugate of p , i.e. p' and p^* satisfy:

$$\frac{1}{p} + \frac{1}{p'} = 1 \quad \text{and} \quad p^* := \frac{Np}{N-p}. \quad (1.1)$$

Note that $p' \in (1, \infty)$ and that $p^* > p$. Finally, Δ_p stands for the p -Laplace operator:

$$\Delta_p u := \operatorname{div}(|\nabla u|^{p-2} \nabla u) = \nabla \cdot (|\nabla u|^{p-2} \nabla u), \quad (1.2)$$

which we interpret in the weak sense.¹

Before proceeding further, we will require some machinery from functional analysis and several simple preliminary tools.

1.1. The Setup. In what follows, $\mathcal{X}$ and $\mathcal{Y}$ each denote normed vector spaces over the real numbers and $A : X \rightarrow Y$ denotes an operator, not necessarily linear. For the sake of convenience, $\mathcal{L}(\mathcal{X}, \mathcal{Y})$ is the real vector space of all bounded linear operators

¹Let U be an open set in $\mathbb{R}^N$ and assume that $u \in W^{1,p}(U)$ is a sufficiently “nice” function satisfying $\Delta_p u = 0$ in U . Testing against $\varphi \in C_c^\infty(U)$, an integration by parts shows that

$$0 = \int_U \varphi \Delta_p u = \int_U \varphi \operatorname{div}(|\nabla u|^{p-2} \nabla u) = - \int_U |\nabla u|^{p-2} \nabla u \cdot \nabla \varphi. \quad (\text{F.1})$$

In light of this, we say that $u \in W^{1,p}(U)$ satisfies $\Delta_p u = 0$ in U if the expression in (F.1) vanishes for all $\varphi \in C_c^\infty(U)$.

$\mathcal{X} \rightarrow \mathcal{Y}$. If $\mathcal{Y}$ is complete, then $\mathcal{L}(\mathcal{X}, \mathcal{Y})$ is a Banach space. Before proceeding further, let us reiterate the notion of a Fréchet derivative.

DEFINITION 1. An operator $A : \mathcal{X} \rightarrow \mathcal{Y}$ is said to be *Fréchet differentiable* at a point $x \in \mathcal{X}$ if there exists a bounded linear operator $DA(x) : \mathcal{X} \rightarrow \mathcal{Y}$ such that

$$\lim_{\substack{h \rightarrow 0 \\ h \in \mathcal{X}}} \frac{\|A(x+h) - A(x) - DA(x)h\|_{\mathcal{Y}}}{\|h\|_{\mathcal{X}}} = 0.$$

The operator $DA(x)$ is called the *Fréchet derivative* of A at x . The *Banach valued* operator

$$DA : \mathcal{X} \rightarrow \mathcal{L}(\mathcal{X}, \mathcal{Y})$$

given by $x \mapsto DA(x)$ is called the *Fréchet derivative* of A , whenever it exists. If this operator exists, we say that A is *Fréchet differentiable* on $\mathcal{X}$. In the case that DA is continuous, we write $A \in C^1(\mathcal{X}, \mathcal{Y})$.

It is not hard to see that a Fréchet differentiable operator $\mathcal{X} \rightarrow \mathcal{Y}$ is automatically continuous. With this in mind, we are ready to formulate the setup of our main result. For N, p, μ and Ω as above, we are interested in solutions to the problem

$$\begin{cases} -\Delta_p u + au^{p-1} \equiv \mu |u|^{p^*-2} u & \text{in } \Omega, \\ u \in W_0^{1,p}(\Omega). \end{cases} \quad (1.3)$$

where $a \in L^{N/p}(\mathbb{R}^N)$ is arbitrary, but fixed. When studying the problem above, solutions to the following *limiting equation* will arise naturally:

$$-\Delta_p u \equiv \mu |u|^{p^*-2} u \quad \text{in } \mathbb{R}^N. \quad (1.4)$$

Next, let us consider the functional on $W_0^{1,p}(\Omega)$ given by:

$$\phi : W_0^{1,p}(\Omega), \quad u \mapsto \int_{\Omega} \left(\frac{|\nabla u|^p}{p} + a \frac{|u|^p}{p} - \mu \frac{|u|^{p^*}}{p^*} \right). \quad (1.5)$$

Through elementary techniques, it can be shown that ϕ is Fréchet differentiable on $W_0^{1,p}(\Omega)$ (and hence continuous on this space) with derivative equal to

$$D\phi(u)(h) = \int_{\Omega} \left(|\nabla u|^{p-2} \nabla u \cdot \nabla h + a |u|^{p-2} uh - \mu |u|^{p^*-2} uh \right), \quad (1.6)$$

for all $u, h \in W_0^{1,p}(\Omega)$. The function ϕ is the energy functional associated with (1.3), and $D\phi$ will be how we test to see whether or not a function $u \in W_0^{1,p}(\Omega)$ satisfies (1.3) in Ω . Much like in (F.1), $u \in W_0^{1,p}(\Omega)$ will be called a *weak solution* to (1.3) if $D\phi(u)$ vanishes on all of $W_0^{1,p}(\Omega)$. To define the energy functional associated to (1.4), we must first introduce the notion of a “homogeneous Sobolev space”, which we accomplish below.

DEFINITION 2. Given an open set $U \subseteq \mathbb{R}^N$, we define $\mathcal{D}^{1,p}(U)$ to be those functions $u \in L^{p^*}(U)$ having weak derivatives in $L^p(U)$. More precisely,

$$\mathcal{D}^{1,p}(U) := \left\{ u \in L^{p^*}(U) : \nabla u \in L^p(\mathbb{R}^N; \mathbb{R}^N) \right\}.$$

We then equip $\mathcal{D}^{1,p}(U)$ with the natural seminorm

$$\|u\| := \|\nabla u\|_{L^p(U)}, \quad \forall u \in \mathcal{D}^{1,p}(U).$$

For general open sets U in $\mathbb{R}^N$, our definition makes $\mathcal{D}^{1,p}(U)$ into a real locally convex topological vector space. When $U = \mathbb{R}^N$, it is easy to see that $\|\cdot\|$ is actually a *norm* on $\mathcal{D}^{1,p}(U)$, and thus $\mathcal{D}^{1,p}(\mathbb{R}^N)$ is a real normed space. Finally, we denote by $\mathcal{D}_0^{1,p}(U)$ the closure of $C_c^\infty(U)$ in $\mathcal{D}^{1,p}(U)$. Note also that $W^{1,p}(\mathbb{R}^N)$ is a subset of $\mathcal{D}^{1,p}(\mathbb{R}^N)$ by the Sobolev embedding theorem.

REMARK 1.1. A priori, there is no reason for $\mathcal{D}^{1,p}(\mathbb{R}^N)$ to be complete. After all, if (u_n) is a Cauchy sequence in $\mathcal{D}^{1,p}(\mathbb{R}^N)$, then (∇u_n) is Cauchy (and thus convergent) in $L^p(\mathbb{R}^N; \mathbb{R}^N)$. However, it is not obvious that this forces (u_n) to also be Cauchy in $L^{p^*}(\mathbb{R}^N)$, nor is it obvious that the would-be $L^{p^*}(\mathbb{R}^N)$ limit of (u_n) is weakly differentiable. Consequently, we encounter some difficulties when trying to show that (u_n) converges to *something* in $\mathcal{D}^{1,p}(\mathbb{R}^N)$, as the $\mathcal{D}^{1,p}(\mathbb{R}^N)$ -norm tests only the convergence of the gradients. But, by using a clever trick from Willem [7], we will later show that $\mathcal{D}^{1,p}(\mathbb{R}^N)$ is in fact a reflexive Banach space.

REMARK 1.2. We must be careful when working with both $W^{1,p}(\mathbb{R}^N)$ and $\mathcal{D}^{1,p}(\mathbb{R}^N)$. As mentioned above, it is true by the Sobolev embedding theorem that $W^{1,p}(\mathbb{R}^N)$ is a subset of $\mathcal{D}^{1,p}(\mathbb{R}^N)$. However, this does not make $W^{1,p}(\mathbb{R}^N)$ a subspace of $\mathcal{D}^{1,p}(\mathbb{R}^N)$. Certainly, the norm on $\mathcal{D}^{1,p}(\mathbb{R}^N)$ tests only the $L^p(\mathbb{R}^N; \mathbb{R}^N)$ -convergence of the gradients, and unfortunately the norm

$$\|u\| := \|\nabla u\|_{L^p(\mathbb{R}^N)}$$

is not equivalent to the $\|\cdot\|_{W^{1,p}(\mathbb{R}^N)}$ norm on $W^{1,p}(\mathbb{R}^N)$. Indeed, this is because the Friedrich-Poincaré inequality fails in $\mathbb{R}^N$. Hence, $W^{1,p}(\mathbb{R}^N)$ inherits a subspace topology from $\mathcal{D}^{1,p}(\mathbb{R}^N)$ different from its canonical topology. However, $W_0^{1,p}(\Omega)$ is a closed subspace of $\mathcal{D}^{1,p}(\mathbb{R}^N)$.

When dealing with the limiting problem (1.4), we will be focused on weak solutions belonging to the space $\mathcal{D}^{1,p}(\mathbb{R}^N)$. To formalize this, we introduce (as above) an energy functional

$$\phi_\infty : \mathcal{D}^{1,p}(\mathbb{R}^N) \rightarrow \mathbb{R}, \quad u \mapsto \int_{\mathbb{R}^N} \left(\frac{|\nabla u|^p}{p} - \mu \frac{|u|^{p^*}}{p^*} \right).$$

This functional is Fréchet differentiable on $\mathcal{D}^{1,p}(\mathbb{R}^N)$ with derivative given by

$$D\phi_\infty(u)(h) = \int_{\mathbb{R}^N} (|\nabla u|^{p-2} \nabla u \cdot \nabla h - \mu |u|^{p^*-2} u h),$$

for $u, h \in \mathcal{D}^{1,p}(\mathbb{R}^N)$. As before, we will say that $u \in \mathcal{D}^{1,p}(\mathbb{R}^N)$ is a weak solution to (1.4) if $D\phi_\infty(u) = 0$ on $\mathcal{D}^{1,p}(\mathbb{R}^N)$. Finally, we will also be treating solutions to the following problem:

$$\begin{cases} -\Delta_p u \equiv \mu |u|^{p^*-2} u & \text{in a halfspace } H, \\ u \in \mathcal{D}_0^{1,p}(H), \end{cases} \quad (1.7)$$

which has a similar weak interpretation.

To state our main result, we need only one last definition.

DEFINITION 3. Let (u_n) be a sequence in $W_0^{1,p}(\Omega)$. We say that (u_n) is a *Palais-Smale* sequence (or simply a P.S.-sequence) for ϕ if (u_n) is bounded, $\phi(u_n)$ converges in $\mathbb{R}$, and

$$D\phi(u_n) \rightarrow 0 \quad \text{in } W^{-1,p'}(\Omega)$$

as $n \rightarrow \infty$. Here, $W^{-1,p'}(\Omega)$ denotes the topological dual of $W_0^{1,p}(\Omega)$ over $\mathbb{R}$. Note that $W^{-1,p'}(\Omega)$ is itself a Banach space.

As it turns out, the requirement that (u_n) be bounded is redundant. It can be shown (see Struwe [5] for a proof in the $p = 2$ case, or Willem [8] for the general case) that any sequence (u_n) belonging to $W_0^{1,p}(\Omega)$ such that $\phi(u_n)$ converges and $D\phi(u_n) \rightarrow 0$ is necessarily bounded. However, since our goal is to illustrate the new method of proof used by Mercuri-Willem, it will not harm us to assume a priori that a Palais-Smale sequence is bounded.

Having given this critical definition, a coherent formulation of our main result (Theorem 2 in Mercuri-Willem [3]) is within reach.

THEOREM 1.1 (Mercuri-Willem). *Let (u_n) be a Palais-Smale sequence for ϕ and assume that the following condition holds:*

$$\inf_{\substack{u \in W_0^{1,p}(\Omega) \\ \|\nabla u\|_{L^p(\Omega)}}} \int_{\Omega} (|\nabla u|^p + a |u|^p) > 0. \quad (\dagger)$$

After possibly passing to a subsequence, there exists a weak solution $v_0 \in W_0^{1,p}(\Omega)$ to the problem (1.3) and a finite (possibly empty) family $v_1, \dots, v_k$ in $\mathcal{D}^{1,p}(\mathbb{R}^N)$ of weak solutions to one of the limiting problems (1.4) or (1.7), with associated sequences $(y_n^i)_{n \in \mathbb{N}} \subset \bar{\Omega}$ and

$(\lambda_n^i)_{n \in \mathbb{N}} \subset \mathbb{R}_+$, such that for all $i = 1, \dots, k$:

$$\left\| u_n - v_0 - \sum_{i=1}^k (\lambda_n^i)^{\frac{p-N}{p}} v_i \left(\frac{\cdot - y_n^i}{\lambda_n^i} \right) \right\| \rightarrow 0 \quad \text{as } n \rightarrow \infty, \quad (1.8)$$

$$\|u_n\|^p \rightarrow \sum_{i=0}^k \|v_i\|^p \quad \text{as } n \rightarrow \infty, \quad (1.9)$$

$$\phi(v_0) + \sum_{i=1}^k \phi_\infty(v_i) = \lim_{n \rightarrow \infty} \phi(u_n). \quad (1.10)$$

REMARK 1.3. This theorem is a very slight modification of Theorem 2 in Mercuri-Willem [3]. In their paper, it was assumed that $(u_n)_- \rightarrow 0$ in $L^p(\mathbb{R}^N)$, as $n \rightarrow \infty$. With this additional hypothesis, they were able to establish the non-negativity of v_i , for every $i = 1, \dots, N$. Moreover, these v_i all solved (1.4) instead of, possibly, (1.7). However, by relaxing the non-negativity assumption in $L^p(\mathbb{R}^N)$ and using a very similar proof to that from Mercuri-Willem [3], we can obtain this slightly more “precise” and inclusive result. However, as will be discussed in the final section, we can drop both the non-negativity assumption and the half-space problem (1.7) in the case $p = 2$.

2. Tools from Measure Theory and Functional Analysis

In this section, we establish several results that will be used freely throughout the proof of Theorem 1.1 and its supporting lemmas. Although not directly related to the problem at hand, they are necessary tools that can be proven using only some elementary concepts from measure theory and functional analysis. We begin with a very useful convergence result reminiscent of the dominated convergence theorem.

THEOREM 2.1 (*p*-Bounded Convergence Theorem). *Let $(X, \mathfrak{M}, \mu)$ be a measure space, $1 < p < \infty$, and let $\{f_n\}_{n \in \mathbb{N}}$ be a bounded family in $L^p(X, \mu)$. Let $f : X \rightarrow \mathbb{C}$ be measurable and assume that*

$$\lim_{n \rightarrow \infty} f_n(x) = f(x)$$

for μ -a.e. $x \in X$. Then, $f \in L^p(X, \mu)$ and

$$\lim_{n \rightarrow \infty} \int_X |(f_n - f)g| d\mu = 0$$

for every $g \in L^{p'}(X, \mu)$. In particular,

$$\lim_{n \rightarrow \infty} \int_X f_n g d\mu = \int_X f g d\mu$$

for all $g \in L^{p'}(X, \mu)$.

PROOF. By Fatou's lemma, we automatically have $f \in L^p(X, \mu)$ since

$$\int_X |f|^p d\mu \leq \liminf_{n \rightarrow \infty} \int_X |f_n|^p d\mu \leq \sup_{n \in \mathbb{N}} \|f_n\|_{L^p(X, \mu)}^p < \infty.$$

Hence, we may assume without loss of generality that $f = 0$ in $L^p(X, \mu)$. Thus, it will sufficient to show that

$$\limsup_{n \rightarrow \infty} \int_X |f_n g| d\mu \leq 0.$$

To this end, let us fix a constant $\alpha > 0$. For $n \in \mathbb{N}$, consider the measurable set

$$E_n := \{x \in X : |f_n(x)g(x)| \leq \alpha |g(x)|^{p'}\}.$$

By the dominated convergence theorem, it is clear that

$$\lim_{n \rightarrow \infty} \int_{E_n} |f_n(x)g(x)| d\mu = \lim_{n \rightarrow \infty} \int_X |f_n(x)g(x)| \mathbf{1}_{E_n}(x) d\mu = 0, \quad (2.1)$$

since $|f_n(x)g(x)| \mathbf{1}_{E_n}(x) \leq \alpha |g(x)|^{p'}$ on X . Let $M > 0$ be such that $\|f_n\|_{L^p(X, \mu)} \leq M$ for every $n \geq 1$. Since

$$|f_n(x)g(x)| > \alpha |g(x)|^{p'}$$

for every $x \in E_n^c$, an application of Hölder's inequality gives

$$\begin{aligned} \int_{E_n^c} |f_n g| d\mu &\leq \|f_n\|_{L^p(E_n^c, \mu)} \left(\int_{E_n^c} |g|^{p'} d\mu \right)^{1/p'} \\ &\leq \alpha^{-1/p'} \|f_n\|_{L^p(X, \mu)} \left(\int_{E_n^c} |f_n g| d\mu \right)^{1/p'} \\ &\leq M \alpha^{-1/p'} \left(\int_{E_n^c} |f_n g| d\mu \right)^{1/p'}. \end{aligned}$$

This yields

$$\left(\int_{E_n^c} |f_n g| d\mu \right)^{1/p} = \left(\int_{E_n^c} |f_n g| d\mu \right)^{1-1/p'} \leq M \alpha^{-1/p'}$$

Or, rather,

$$\int_{E_n^c} |f_n g| d\mu \leq M^p \alpha^{-p/p'}, \quad \forall n \in \mathbb{N}.$$

Combining this with (2.1), it follows that

$$\limsup_{n \rightarrow \infty} \int_X |f_n g| d\mu \leq \limsup_{n \rightarrow \infty} \int_{E_n^c} |f_n g| d\mu \leq M^p \alpha^{-p/p'}.$$

Finally, sending $\alpha \rightarrow \infty$ verifies the assertion. $\square$

This gives us the following very useful convergence criterion, which can be thought of as a “local version” of Theorem 2.1.

COROLLARY 2.1. *Let $(X, \mathfrak{M}, \mu)$ be a measure space and fix $1 < p < \infty$. Let $f : X \rightarrow \mathbb{C}$ be a measurable function and assume that $\{f_n\}_{n \in \mathbb{N}} \subseteq L^p_{\text{loc}}(X, \mu)$ is such that*

$$\lim_{n \rightarrow \infty} f_n(x) = f(x)$$

for μ -a.e. $x \in X$. If

$$\sup_{n \in \mathbb{N}} \int_E |f_n|^p d\mu < \infty$$

for every $E \in \mathfrak{M}$ having finite measure, then $f_n \rightarrow f$ in $L^1_{\text{loc}}(X, \mu)$ as $n \rightarrow \infty$.

LEMMA 2.2. *Let (X, d) be a compact metric space and denote by $C(X)$ the real vector space of all continuous maps $X \rightarrow \mathbb{R}$, endowed with the norm*

$$\|\cdot\|_\infty := \sup_{x \in X} |\cdot(x)| < \infty.$$

If $\mathcal{F} \subseteq C(X)$ is equicontinuous and equibounded, then

$$\sup_{f \in \mathcal{F}} f \in C(X).$$

Namely, $\sup_{f \in \mathcal{F}} f$ is continuous on X .

PROOF. First, we define a function $s : X \rightarrow \mathbb{R}$ via the rule

$$s(x) := \sup_{f \in \mathcal{F}} f(x);$$

since the family $\mathcal{F}$ is equibounded, this is clearly a well defined function. Next, consider any two points $x, y \in X$. Clearly,

$$\sup_{f \in \mathcal{F}} f(x) = \sup_{f \in \mathcal{F}} (f(x) - f(y) + f(y)) \leq \sup_{f \in \mathcal{F}} (f(x) - f(y)) + \sup_{f \in \mathcal{F}} f(y)$$

whence

$$\sup_{f \in \mathcal{F}} f(x) - \sup_{f \in \mathcal{F}} f(y) \leq \sup_{f \in \mathcal{F}} (f(x) - f(y)) \leq \sup_{f \in \mathcal{F}} |f(x) - f(y)|.$$

Similarly, we have

$$\sup_{f \in \mathcal{F}} f(y) - \sup_{f \in \mathcal{F}} f(x) \leq \sup_{f \in \mathcal{F}} (f(y) - f(x)) \leq \sup_{f \in \mathcal{F}} |f(x) - f(y)|.$$

Combining these two inequalities yields

$$\left| \sup_{f \in \mathcal{F}} f(x) - \sup_{f \in \mathcal{F}} f(y) \right| \leq \sup_{f \in \mathcal{F}} |f(x) - f(y)|.$$

Let now $\varepsilon > 0$ be given. By equicontinuity, there is $\delta > 0$ such that $|f(x) - f(y)| < \varepsilon$ for all $f \in \mathcal{F}$, whenever $d(x, y) < \delta$. For any such $x, y \in X$, we must therefore have

$$\left| \sup_{f \in \mathcal{F}} f(x) - \sup_{f \in \mathcal{F}} f(y) \right| \leq \sup_{f \in \mathcal{F}} |f(x) - f(y)| \leq \sup_{f \in \mathcal{F}} \varepsilon = \varepsilon.$$

This shows that $s(x)$ is continuous and the proof is complete. $\square$

Our only application of this lemma will be the following.

PROPOSITION 2.3. *Let $U \subseteq \mathbb{R}^N$ be non-empty and let $f \in L^1(\mathbb{R}^N)$. For $r \geq 0$, we define the Lévy concentration function of f as*

$$Q(r) := \sup_{y \in U} \int_{B(y, r)} |f|. \quad (2.2)$$

The function $Q(r)$ is continuous on $[0, \infty)$.

PROOF. For fixed $y \in U$, we define

$$Q_y : [0, \infty) \rightarrow \mathbb{R}, \quad r \mapsto \int_{B(y, r)} |f|.$$

Since $f \in L^1(\mathbb{R}^N)$, each Q_y is continuous at 0, i.e.

$$\lim_{r \searrow 0} Q_y(r) = 0.$$

By virtue of Lemma 2.2, it is enough to show that the family $\mathcal{F} := \{Q_y\}_{y \in U}$ is equicontinuous and equibounded on any compact interval $[a, b]$. To see that this family is equibounded, let $y \in U$ and $r \geq 0$ be arbitrary. Clearly, one has the following:

$$|Q_y(r)| = \int_{B(y, r)} |f| \leq \int_{\mathbb{R}^N} |f| < \infty.$$

It follows that $\mathcal{F}$ is equibounded on $[0, \infty)$, and hence on $[a, b]$. Next, we establish the equicontinuity of $\mathcal{F}$ on $[a, b]$. Let $\varepsilon > 0$ be given. Since $f \in L^1(\mathbb{R}^N)$, there exists $\delta > 0$ such that

$$\int_E |f| < \varepsilon$$

whenever $E \subset \mathbb{R}^N$ is Lebesgue measurable with $m(E) < \delta$.² Now, for any $r, q \in [a, b]$ and every $y \in U$, it is easily seen that

$$|Q_y(r) - Q_y(q)| = \left| \int_{B(y, r)} |f| - \int_{B(y, q)} |f| \right| = \int_{B(y, r) \setminus B(y, q)} |f|$$

²Here, $m(\cdot)$ denotes the Lebesgue measure on $\mathbb{R}^N$.

where we are assuming, without loss of generality, that $r \geq q$. Letting ω_n be the volume of the unit ball in $\mathbb{R}^N$, we have

$$\begin{aligned} m(B(y, r) \setminus B(y, q)) &= \omega_n(r^n - q^n) \\ &= \omega_n(r^{n-1} + r^{n-2}q + \cdots + rq^{n-2} + q^{n-1})(r - q) \\ &\leq C(r - q), \end{aligned}$$

for a constant $C > 0$ independent of $r, q \in [a, b]$.³ Therefore, we will have

$$m(B(y, r) \setminus B(y, q)) < \delta$$

whenever $0 \leq (r - q) < \frac{\delta}{C}$. This implies that

$$|Q_y(r) - Q_y(q)| \leq \int_{B(y, r) \setminus B(y, q)} |f| < \varepsilon$$

for any such $r, q \in [a, b]$. This shows that the family $\mathcal{F}$ is equicontinuous and equibounded on every $[a, b] \subset [0, \infty)$ and this completes the proof. $\square$

2.1. The Topological Structure of $\mathcal{D}^{1,p}(\mathbb{R}^N)$. Let us now discuss the homogeneous space $\mathcal{D}^{1,p}(\mathbb{R}^N)$, which will be of particular interest to us. We have already given this space a norm:

$$\|u\| := \|\nabla u\|_{L^p(\mathbb{R}^N)},$$

but we may also give it the norm:

$$\|u\|_* := \|u\|_{L^{p^*}(\mathbb{R}^N)} + \|\nabla u\|_{L^p(\mathbb{R}^N)} = \|u\|_{L^{p^*}(\mathbb{R}^N)} + \|u\|.$$

Recalling the proof that $W^{1,p}(\mathbb{R}^N) = W_0^{1,p}(\mathbb{R}^N)$, it becomes clear that $C_c^\infty(\mathbb{R}^N)$ is dense in $\mathcal{D}^{1,p}(\mathbb{R}^N)$ with respect to the norm $\|\cdot\|_*$. Now, we have the Gagliardo-Nirenberg-Sobolev inequality:

$$\|\varphi\|_{L^{p^*}} \leq C \|\nabla \varphi\|_{L^p(\mathbb{R}^N)} \quad (2.3)$$

for all $\varphi \in C_c^\infty(\mathbb{R}^N)$, and a constant $C > 0$ independent of φ . Let $u \in \mathcal{D}^{1,p}(\mathbb{R}^N)$ be given and choose a sequence (φ_n) in $C_c^\infty(\mathbb{R}^N)$ such that

$$\lim_{n \rightarrow \infty} \|u - \varphi_n\|_* = 0.$$

Then, we have both

$$\lim_{n \rightarrow \infty} \|u - \varphi_n\|_{L^{p^*}(\mathbb{R}^N)} = 0 \quad \text{and} \quad \lim_{n \rightarrow \infty} \|\nabla u - \nabla \varphi_n\|_{L^p(\mathbb{R}^N; \mathbb{R}^N)} = 0.$$

By continuity of norms, this combined with (2.3) gives

$$\begin{aligned} \|u\|_{L^{p^*}(\mathbb{R}^N)} &= \lim_{n \rightarrow \infty} \|\varphi_n\|_{L^{p^*}} \leq C \lim_{n \rightarrow \infty} \|\nabla \varphi_n\|_{L^p(\mathbb{R}^N; \mathbb{R}^N)} \\ &\leq C \|\nabla u\|_{L^p(\mathbb{R}^N)}. \end{aligned}$$

³Note that this constant C will depend on the interval $[a, b]$.

In other words, the Gagliardo-Nirenberg-Sobolev inequality holds for all $u \in \mathcal{D}^{1,p}(\mathbb{R}^N)$. In fact, this shows that $\|\cdot\|$ is equivalent to the norm $\|\cdot\|_*$ on $\mathcal{D}^{1,p}(\mathbb{R}^N)$. Indeed, for any $u \in \mathcal{D}^{1,p}(\mathbb{R}^N)$ there holds

$$\begin{aligned} \|u\| &= \|\nabla u\|_{L^p(\mathbb{R}^N)} \leq \|\nabla u\|_{L^p(\mathbb{R}^N)} + \|u\|_{L^{p^*}(\mathbb{R}^N)} \\ &\leq C \|\nabla u\|_{L^p(\mathbb{R}^N)} \\ &= C \|u\| \end{aligned}$$

with $C > 0$ a constant depending only on N and p . We summarize our findings in the following theorem:

THEOREM 2.2 (Properties of $\mathcal{D}^{1,p}(\mathbb{R}^N)$). *If $1 < p < \infty$, then*

- (1) $\mathcal{D}^{1,p}(\mathbb{R}^N)$ is a Banach space;
- (2) smooth functions of compact support are dense in $\mathcal{D}^{1,p}(\mathbb{R}^N)$;
- (3) the Gagliardo-Nirenberg-Sobolev inequality holds in $\mathcal{D}^{1,p}(\mathbb{R}^N)$:

$$\|u\|_{L^{p^*}} \leq C \|\nabla u\|_{L^p(\mathbb{R}^N)}, \quad \forall u \in \mathcal{D}^{1,p}(\mathbb{R}^N); \quad (\text{GNS})$$

- (4) $\mathcal{D}^{1,p}(\mathbb{R}^N) \subseteq W_{loc}^{1,p}(\mathbb{R}^N)$.

In particular, bounded sequences in $\mathcal{D}^{1,p}(\mathbb{R}^N)$ are bounded in $L^{p^}(\mathbb{R}^N)$.*

PROOF. By our previous discussion, we need only verify the first point. In fact, it will be enough to show that $\mathcal{D}^{1,p}(\mathbb{R}^N)$ is complete with respect to the norm $\|\cdot\|_*$. To this end, we fix a Cauchy sequence (u_n) in $(\mathcal{D}^{1,p}(\mathbb{R}^N), \|\cdot\|_*)$. Clearly, (u_n) is Cauchy in $L^{p^*}(\mathbb{R}^N)$ and $(\partial_i u_n)$ is Cauchy in $L^p(\mathbb{R}^N)$ for each $i = 1, \dots, N$. Thus, we may find $u_0 \in L^{p^*}(\mathbb{R}^N)$ and $u_1, \dots, u_N \in L^p(\mathbb{R}^N)$ such that $\lim_{n \rightarrow \infty} u_n = u_0$ in $L^{p^*}(\mathbb{R}^N)$ and

$$u_i = \lim_{n \rightarrow \infty} \partial_i u_n \quad \text{in } L^p(\mathbb{R}^N),$$

for every $i = 1, \dots, N$. By definition of $\|\cdot\|_*$, the claim will follow if $\partial_i u_0 = u_i$ for each index i . Let $\varphi \in C_c^\infty(\mathbb{R}^N)$ be given; two applications of Hölder's inequality shows that

$$\begin{aligned} \int_{\mathbb{R}^N} u_0 \partial_i \varphi &= \lim_{n \rightarrow \infty} \int_{\mathbb{R}^N} u_n \partial_i \varphi = - \lim_{n \rightarrow \infty} \int_{\mathbb{R}^N} \varphi \partial_i u_n \\ &= - \int_{\mathbb{R}^N} \varphi u_i. \end{aligned}$$

This establishes the assertion. □

2.2. Understanding Weak Convergence in $\mathcal{D}^{1,p}(\mathbb{R}^N)$. As mentioned previously, the dual space of $\mathcal{D}^{1,p}(\mathbb{R}^N)$ will play an important role in the proof of Theorem 1.1. Consequently, we seek to properly understand weak convergence in $\mathcal{D}^{1,p}(\mathbb{R}^N)$. Of course, since strong convergence in $\mathcal{D}^{1,p}(\mathbb{R}^N)$ only tests the L^p -convergence of the gradients, we

should expect that weak convergence in $\mathcal{D}^{1,p}(\mathbb{R}^N)$ be characterized by the weak convergence of the gradients in L^p . As it turns out, this is indeed the case.

PROPOSITION 2.4. *Let (v_n) be a sequence in $\mathcal{D}^{1,p}(\mathbb{R}^N)$ and fix $v \in \mathcal{D}^{1,p}(\mathbb{R}^N)$. Then, $v_n \rightharpoonup v$ in $\mathcal{D}^{1,p}(\mathbb{R}^N)$ if and only if $\nabla v_n \rightharpoonup \nabla v$ in $L^p(\mathbb{R}^N; \mathbb{R}^N)$.*

PROOF. We begin by defining the following map

$$T : \mathcal{D}^{1,p}(\mathbb{R}^N) \rightarrow L^p(\mathbb{R}^N; \mathbb{R}^N), \quad u \mapsto \nabla u.$$

Clearly, T is a linear isometry and is hence an embedding. Through T , we can identify $\mathcal{D}^{1,p}(\mathbb{R}^N)$ with the subspace $T(\mathbb{R}^N)$ of $L^p(\mathbb{R}^N; \mathbb{R}^N)$. As a first step, we will show that $v_n \rightharpoonup v$ in $\mathcal{D}^{1,p}(\mathbb{R}^N)$ if and only if $\nabla v_n \rightharpoonup \nabla v$ in $T(\mathbb{R}^N)$. Let us now assume that v_n converges weakly to v in $\mathcal{D}^{1,p}(\mathbb{R}^N)$. Given a continuous linear functional ϕ on $T(\mathbb{R}^N)$, we define

$$\varphi : \mathcal{D}^{1,p}(\mathbb{R}^N) \rightarrow \mathbb{R}, \quad \varphi(u) := \phi(\nabla u).$$

In other words, we put $\varphi := \phi \circ T$. Clearly, φ is a continuous linear functional on $\mathcal{D}^{1,p}(\mathbb{R}^N)$. Since v_n converges weakly to v in $\mathcal{D}^{1,p}(\mathbb{R}^N)$, we must have $\varphi(v_n) \rightarrow \varphi(v)$ as $n \rightarrow \infty$. However, this is equivalent to the statement

$$\lim_{n \rightarrow \infty} \phi(\nabla v_n) = \phi(\nabla v).$$

This shows that ∇v_n converges weakly to ∇v , in $L^p(\mathbb{R}^N; \mathbb{R}^N)$, as $n \rightarrow \infty$. Conversely, we assume that $\nabla v_n \rightharpoonup \nabla v$ in $T(\mathbb{R}^N)$, and fix a continuous linear functional ϕ on $\mathcal{D}^{1,p}(\mathbb{R}^N)$. Next, consider the map

$$\phi : T(\mathbb{R}^N) \rightarrow \mathbb{R}, \quad \phi := \varphi \circ T^{-1}.$$

Again, it is obvious that ϕ is a continuous linear functional on $T(\mathbb{R}^N)$. Since ∇v_n converges weakly to ∇v in $T(\mathbb{R}^N)$, we see that $\phi(\nabla v_n)$ converges to $\phi(\nabla v)$. By definition, this is equivalent to having

$$\lim_{n \rightarrow \infty} \varphi(v_n) = \varphi(v).$$

To summarize, we have shown that weak convergence in $\mathcal{D}^{1,p}(\mathbb{R}^N)$ is equivalent to the weak convergence of the gradients in $T(\mathbb{R}^N)$. To complete the proof, we need only show that weak convergence of the gradients in $T(\mathbb{R}^N)$ is equivalent to their weak convergence in all of $L^p(\mathbb{R}^N)$. Here, the only non-trivial claim is that weak convergence in $L^p(\mathbb{R}^N; \mathbb{R}^N)$ implies weak convergence in $T(\mathbb{R}^N)$. However, this is a simple consequence of the Hahn-Banach theorem. Indeed, any continuous linear functional ϕ on $T(\mathbb{R}^N)$ can be extended to a continuous linear functional ψ on all of $L^p(\mathbb{R}^N; \mathbb{R}^N)$. Thus, if $\nabla v_n \rightharpoonup \nabla v$ in $L^p(\mathbb{R}^N; \mathbb{R}^N)$, then

$$\psi(v_n) \rightarrow \psi(v) \quad \text{in } L^p(\mathbb{R}^N; \mathbb{R}^N) \text{ as } n \rightarrow \infty.$$

Clearly, this simply means that $\lim_{n \rightarrow \infty} \phi(v_n) = \phi(v)$ whence we have established that ∇v_n converges weakly to ∇v in $T(\mathbb{R}^N)$. This completes the proof. $\square$

This technical result will mainly be used when establishing the following:

COROLLARY 2.5. *If (v_n) is a bounded sequence in $\mathcal{D}^{1,p}(\mathbb{R}^N)$ there exists $v \in \mathcal{D}^{1,p}(\mathbb{R}^N)$ and a subsequence of (v_n) that converges weakly to v in $\mathcal{D}^{1,p}(\mathbb{R}^N)$.*

REMARK 2.1. Note that we cannot directly apply any familiar results about weak compactness in Banach spaces as $\mathcal{D}^{1,p}(\mathbb{R}^N)$ may not be reflexive, a priori. However, as will become apparent in the proof, we will be “saved” by the familiar properties of L^p -spaces. Indeed, the requirement $p \in (1, \infty)$ implies that $p' \in (1, \infty)$. Therefore, duality tells us that

$$(L^p(\mathbb{R}^N; \mathbb{R}^N))^{**} \cong (L^{p'}(\mathbb{R}^N; \mathbb{R}^N))^* \cong L^p(\mathbb{R}^N; \mathbb{R}^N)$$

whence we see that $L^p(\mathbb{R}^N; \mathbb{R}^N)$ is reflexive. If $\mathcal{X}$ is a Banach space, then $\mathcal{X}$ is reflexive if and only if the closed unit ball in $\mathcal{X}$ is weakly compact⁴. It therefore follows that bounded sequences in $L^p(\mathbb{R}^N; \mathbb{R}^N)$ have a weakly convergent subsequence. Since $W^{1,p}(\Omega)$ is reflexive for $1 < p < \infty$, the reasoning shows that bounded sequences in $W^{1,p}(\Omega)$ have weakly convergent subsequences.

PROOF OF COROLLARY. By definition of the $\mathcal{D}^{1,p}(\mathbb{R}^N)$ -norm, the sequence of gradients (∇v_n) is bounded in $L^p(\mathbb{R}^N; \mathbb{R}^N)$. By Theorem 2.2, the sequence (v_n) is also bounded in $L^{p^*}(\mathbb{R}^N)$. By basic L^p -theory, after passing to a subsequence, we may assume that ∇v_n converges weakly to some $w = (w_1, \dots, w_N)$ in $L^p(\mathbb{R}^N; \mathbb{R}^N)$. Passing to yet another subsequence, we may also assume that v_n converges weakly to $v \in L^{p^*}(\mathbb{R}^N)$. Next, we claim that v belongs to $\mathcal{D}^{1,p}(\mathbb{R}^N)$ and that $\nabla v = w$. To see this, fix an index $i = 1, \dots, N$ and let $\varphi \in C_c^\infty(\mathbb{R}^N)$ be given. Weak convergence in $L^{p^*}(\mathbb{R}^N)$ gives

$$\int_{\mathbb{R}^N} v \partial_i \varphi = \lim_{n \rightarrow \infty} \int_{\mathbb{R}^N} v_n \partial_i \varphi = - \lim_{n \rightarrow \infty} \int_{\mathbb{R}^N} \varphi \partial_i v_n.$$

Since ∇v_n converges weakly to w in $L^p(\mathbb{R}^N; \mathbb{R}^N)$, it is not hard to check that also $\partial_i v_n$ converges weakly to w_i in $L^p(\mathbb{R}^N)$. Using this with the above, we discover that

$$\int_{\mathbb{R}^N} v \partial_i \varphi = - \lim_{n \rightarrow \infty} \int_{\mathbb{R}^N} \varphi \partial_i v_n = - \int_{\mathbb{R}^N} \varphi w_i.$$

We conclude that $v \in \mathcal{D}^{1,p}(\mathbb{R}^N)$ and that $\nabla v = w$. Finally, since $\nabla v_n \rightarrow w = \nabla v$ weakly in $L^p(\mathbb{R}^N; \mathbb{R}^N)$, it follows from the previous lemma that v_n converges weakly to v in $\mathcal{D}^{1,p}(\mathbb{R}^N)$. $\square$

COROLLARY 2.6. *For $1 < p < \infty$, the homogeneous Sobolev space $\mathcal{D}^{1,p}(\mathbb{R}^N)$ is a reflexive Banach space.*

Before starting the proof of Theorem 1.1, we would like to partially strengthen the conclusion of this corollary. Obviously, it would be ideal to extract a subsequence that

⁴The proof of this fact is essentially an application of the Banach-Alaoglu theorem.

converges strongly in $\mathcal{D}^{1,p}(\mathbb{R}^N)$. Instead, we will have to be content with the following pointwise result.

COROLLARY 2.7. *In the context of Corollary 2.5 and its proof, there is a further subsequence that converges pointwise almost everywhere in $\mathbb{R}^N$ to v .*

PROOF. For each $k \in \mathbb{N}$ we denote by B_k the open ball $B(0, k)$ in $\mathbb{R}^N$. For a fixed k , we consider the restrictions of the sequence (v_n) to B_k . In doing so, we obtain a bounded sequence in $W^{1,p}(B_k)$, for each $k \in \mathbb{N}$. By the Rellich-Kondrachov theorem, the embedding

$$W^{1,p}(B_k) \hookrightarrow L^p(B_k)$$

is compact. Thus, passing to a subsequence if necessary, we may assume that

- (i) v_n converges weakly to some $u_k \in W^{1,p}(B_k)$ as $n \rightarrow \infty$;
- (ii) v_n converges *strongly* to u_k in $L^p(B_k)$ as $n \rightarrow \infty$; and
- (iii) v_n converges pointwise almost everywhere to u_k on B_k as $n \rightarrow \infty$.

By duality and the fact that $v_n \rightharpoonup v$ in $\mathcal{D}^{1,p}(\mathbb{R}^N)$, we have by Proposition 2.4 that

$$\lim_{n \rightarrow \infty} \int_{\mathbb{R}^N} g \cdot \nabla v_n = \int_{\mathbb{R}^N} g \cdot \nabla v,$$

for all $g \in L^{p'}(\mathbb{R}^N; \mathbb{R}^N)$. In particular,

$$\lim_{n \rightarrow \infty} \int_{B_k} g \cdot \nabla v_n = \int_{B_k} g \cdot \nabla v$$

for all $g \in L^p(B_k; \mathbb{R}^N)$. On the other hand, (i) ensures that

$$\lim_{n \rightarrow \infty} \int_{B_k} \nabla v_n \cdot g = \lim_{n \rightarrow \infty} \int_{B_k} \nabla u_k \cdot g$$

for all $g \in L^{p'}(B_k)$. Combining these last two equations shows that

$$\int_{B_k} (\nabla u_k - \nabla v) \cdot g = \lim_{n \rightarrow \infty} \int_{B_k} (\nabla u_k - \nabla v_n) \cdot g = 0$$

for all such g . By density, it follows that $u_k = v + \alpha_k$ on B_k for some $\alpha_k \in \mathbb{R}$. In other words, v_k converges pointwise almost everywhere to $v + \alpha_k$ on B_k , as $n \rightarrow \infty$. To complete the proof, we will now make use of a diagonal argument. For this family of balls $\{B_k\}_{k \geq 1}$, we inductively construct subsequences $\{v_{n,k}\}_{k \geq 1}$ for which the above holds true. Therefore, there is a corresponding sequence $(\alpha_k)_{k \geq 1}$ of real numbers such that

$$v_{n,k} \rightarrow v + \alpha_k \quad \text{pointwise a.e. on } B_k \text{ as } n \rightarrow \infty,$$

for each k . Finally, for each $k \in \mathbb{N}$, an application of Fatou's lemma yields

$$\begin{aligned} \alpha_k m(B_k)^{p^*} &\leq \liminf_{n \rightarrow \infty} \int_{B_k} |v_{n,k} - v|^{p^*} \leq \limsup_{n \rightarrow \infty} \int_{B_k} (|v_{n,k}| + |v|)^{p^*} \\ &\leq 2^{p^*-1} \limsup_{n \rightarrow \infty} \left(\int_{B_k} |v_{n,k}|^{p^*} + \int_{B_k} |v|^{p^*} \right) \\ &\leq 2^{p^*-1} \sup_{n \in \mathbb{N}} \left(\int_{\mathbb{R}^N} |v_{n,k}|^{p^*} + \int_{\mathbb{R}^N} |v|^{p^*} \right) \end{aligned}$$

where this last term is uniformly bounded in n, k since (v_n) is bounded in $\mathcal{D}^{1,p}(\mathbb{R}^N)$ and $v \in L^{p^*}(\mathbb{R}^N)$. Noting that $m(B_k) \rightarrow \infty$ as $k \rightarrow \infty$, we conclude that $\alpha_k \rightarrow 0$ as $k \rightarrow \infty$. Then, the diagonal subsequence $(v_{n,n})_{n \in \mathbb{N}}$ will converge pointwise almost everywhere to v on $\mathbb{R}^N$ as $n \rightarrow \infty$, which proves the claim. $\square$

3. Preliminary Results

As mentioned above, we begin by stating two results established by Mercuri-Willem that will play a critical role in the proof of Theorem 1.1. The first is a rather technical, but elementary, identity:

LEMMA A (Lemma 3.2 in [3]). *Let $1 < q < \infty$ and consider the function*

$$A : \mathbb{R}^N \rightarrow \mathbb{R}^N, \quad y \mapsto y |y|^{q-2}.$$

Let μ be a measure on Ω and assume that (u_n) is a bounded sequence in $L^r(\Omega, \mu)$ having the property that

$$\lim_{n \rightarrow \infty} u_n(x) = u(x)$$

for μ -a.e. $x \in \Omega$. Then,

$$\lim_{n \rightarrow \infty} \int_{\Omega} |A(u_n) - A(u_n - u) - A(u)|^{\frac{q}{q-1}} d\mu = 0.$$

THEOREM B (Theorem 3.3 in [3]). *Let (Ω_k) be a sequence of open sets such that $\Omega_k \nearrow \Omega$. Assume that $q > 1$ and (v_n) is a sequence in $W^{1,q}(\Omega)$ such that $v_n \rightharpoonup v$ in $W^{1,q}(\Omega)$. Let $T : \mathbb{R} \rightarrow \mathbb{R}$ be given by the rule*

$$T(s) := \begin{cases} s & \text{if } |s| \leq 1, \\ \frac{s}{|s|} & \text{if } |s| > 1. \end{cases}$$

If for each $k \geq 1$ there holds

$$\lim_{n \rightarrow \infty} \int_{\Omega_k} (|\nabla v_n|^{q-2} \nabla v_n - |\nabla v|^{q-2} \nabla v) \cdot \nabla T(v_n - v) = 0,$$

then, after passing to a subsequence,

- (1) $\nabla v_n \rightarrow \nabla v$ pointwise almost everywhere in Ω ;
- (2) $\lim_{n \rightarrow \infty} \left(\|\nabla v_n\|_{L^q(\Omega)}^q - \|\nabla(v_n - v)\|_{L^q(\Omega)}^q \right) = \|\nabla v\|_{L^q(\Omega)}^q$; and
- (3) $|\nabla v_n|^{q-2} \nabla v_n - |\nabla(v_n - v)|^{q-2} \nabla(v_n - v) \rightarrow |\nabla v|^{q-2} \nabla v$ in $L^{\frac{p}{p-1}}(\Omega)$.

Furthermore, the theorem holds when $\Omega = \mathbb{R}^N$ and $W^{1,q}(\Omega)$ is replaced by $\mathcal{D}^{1,p}(\mathbb{R}^N)$.

Finally, we state a theorem which does not directly pertain to our proof, but will nonetheless lead to a notable remark.

THEOREM C. Any non-negative weak solution $u \in \mathcal{D}_0^{1,p}(\mathbb{R}_+^N)$ to the problem

$$-\Delta_p u \equiv \mu u^{p^*-1} \quad \text{in } \mathbb{R}_+^N$$

vanishes almost everywhere.

3.1. Technical Lemmas. This subsection is devoted to establishing two special lemmas that can be viewed as the “base cases” for the iterative argument that will be used in the proof of Theorem 1.1.

LEMMA 3.1. Suppose that $(\dagger)$ holds true and let (u_n) be a sequence in $W_0^{1,p}(\Omega)$. In addition, assume that each of the following hold true:

- (1) u_n converges weakly to u in $W_0^{1,p}(\Omega)$;
- (2) $u_n \rightarrow u$ pointwise almost everywhere on Ω ;
- (3) $\phi(u_n) \rightarrow c \in \mathbb{R}$ as $n \rightarrow \infty$; and
- (4) $D\phi(u_n) \rightarrow 0$ in $W^{-1,p'}(\Omega)$ as $n \rightarrow \infty$.

After passing to a subsequence, one has $\nabla u_n \rightarrow \nabla u$ pointwise almost everywhere on Ω and $D\phi(u) = 0$. Furthermore, by defining $v_n := u_n - u$ we obtain a sequence in $W_0^{1,p}(\Omega)$ with the property that

- (i) $\lim_{n \rightarrow \infty} (\|u_n\|^p - \|v_n\|^p) = \|u\|^p$;
- (ii) $\lim_{n \rightarrow \infty} \phi_\infty(v_n) = \lim_{n \rightarrow \infty} \phi(u_n) - \phi(u)$; and
- (iii) $D\phi_\infty(v_n) \rightarrow 0$ in $W^{-1,p'}(\Omega)$ as $n \rightarrow \infty$.

PROOF. We follow, with only slight modifications, the proof from Mercuri-Willem [3]. Define $T : \mathbb{R}^N \rightarrow \mathbb{R}^N$ as in Theorem B. Since Ω is bounded and $|T| \leq 1$ on $\mathbb{R}$, the dominated convergence theorem implies that

$$\lim_{n \rightarrow \infty} \int_{\Omega} |T(u_n - u)|^q = 0 \tag{3.1}$$

for all $0 < q < \infty$. Here, we have used the continuity of T and the assumption that $u_n \rightarrow u$ pointwise almost everywhere on Ω . Next, we claim that $T(u_n - u) \rightarrow 0$ in $W_0^{1,p}(\Omega)$. Of course, this amounts to showing that

$$T(v_n) \rightarrow 0 \quad \text{in } W_0^{1,p}(\Omega).$$

Note that this is well defined (see Ziemer [9]) because T is Lipschitz continuous with $T(0) = 0$. Since v_n is weakly convergent, it is bounded in $W_0^{1,p}(\Omega)$. By compactness of the embedding $W_0^{1,p}(\Omega) \hookrightarrow L^p(\Omega)$, we may assume that (v_n) converges strongly to some function in $L^p(\Omega)$. However, since $u_n \rightarrow u$ pointwise almost everywhere in Ω , we must have $v_n \rightarrow 0$ strongly in $L^p(\Omega)$. Clearly, $(T(v_n))$ is bounded in $L^p(\Omega)$. By the chain-rule (which is valid because T is Lipschitz continuous – refer to Ziemer [9]), one has

$$\partial_i[T(v_n)] = T'(v_n)\partial_i v_n$$

for all $i = 1, \dots, N$. Since the derivative of a Lipschitz continuous function is bounded, it follows that $(\nabla T(v_n))$ is bounded in $L^p(\Omega)$ whence $(T(v_n))$ is bounded in $W_0^{1,p}(\Omega)$ as well. After passing to yet another subsequence, it is of no harm to assume that

$$T(v_n) \rightharpoonup \eta \quad \text{in } W_0^{1,p}(\Omega)$$

as $n \rightarrow \infty$. In fact, by compactness of the embedding $W_0^{1,p}(\Omega) \hookrightarrow L^p(\Omega)$, we might as well assume that $T(v_n) \rightarrow \eta$ *strongly* in $L^p(\Omega)$ and pointwise almost everywhere on Ω . Finally, since $u_n \rightarrow u$ pointwise almost everywhere on Ω , we have

$$\lim_{n \rightarrow \infty} T(v_n(x)) = 0 \quad \text{a.e. on } \Omega.$$

This means that $\eta = 0$ for almost every $x \in \Omega$. In particular, $T(v_n)$ converges weakly to 0 in $W^{1,p}(\Omega)$. Next, direct computation shows that

$$\begin{aligned} \int_{\Omega} (|\nabla u_n|^{p-2} \nabla u_n - |\nabla u_n^{p-2}| \nabla u) \cdot \nabla T(u_n - u) &= D\phi(u_n)(T(u_n - u)) \\ &\quad - \underbrace{\int_{\Omega} |\nabla u|^{p-2} \nabla u \cdot \nabla T(u_n - u)}_{I_1} \\ &\quad - \underbrace{\int_{\Omega} (a(x) |u_n|^{p-2} u_n - \mu |u_n|^{p^*-2} u_n) T(u_n - u)}_{I_2}. \end{aligned}$$

Since $D\phi(u_n) \rightarrow 0$ strongly in $W^{-1,p'}(\Omega)$ and $T(u_n - u)$ is bounded in $W_0^{1,p}(\Omega)$, it is easy to see that $D\phi(u_n)(T(u_n - u)) \rightarrow 0$ as $n \rightarrow \infty$. To deal with I_1 , note that by Hölder's inequality,

$$\begin{aligned} |I_1| &\leq \int_{\Omega} |\nabla u|^{p-1} |\nabla T(u_n - u)| \leq \left(\int_{\Omega} |\nabla u|^{(p-1) \cdot \frac{p}{p-1}} \right)^{1/p'} \|\nabla T(u_n - u)\|_{L^p(\Omega)} \\ &\leq \|u\|_{W^{1,p}(\Omega)} \|T(u_n - u)\|_{W^{1,p}(\Omega)}. \end{aligned}$$

Thus, since $T(u_n - u) \rightarrow 0$ in $W_0^{1,p}(\Omega)$, we have $I_1 \rightarrow 0$ as $n \rightarrow \infty$. For convenience, we now break I_2 into two parts. Again, Hölder's inequality gives

$$\begin{aligned} \left| \int_{\Omega} \mu |u_n|^{p^*-2} u_n T(u_n - u) \right| &\leq \mu \int_{\Omega} |u_n|^{p^*-1} |T(u_n - u)| \\ &\leq \mu \left(\int_{\Omega} |u_n|^{(p^*-1) \cdot \frac{p^*}{p^*-1}} \right)^{\frac{p^*-1}{p^*}} \|T(u_n)\|_{L^p(\Omega)} \\ &= \mu \left(\int_{\Omega} |u_n|^{p^*} \right)^{\frac{p^*-1}{p^*}} \|T(u_n)\|_{L^p(\Omega)} \xrightarrow{n \rightarrow \infty} 0, \end{aligned}$$

where we have used here the boundedness of (u_n) in $L^{p^*}(\Omega)$ and (3.1). Next, Hölder's inequality shows that

$$\int_{\Omega} |a| |u_n|^{p-1} |T(u_n - u)| \leq \|a\|_{L^{N/p}(\Omega)} \left(\int_{\Omega} (|u_n|^{p-1} |T(u_n - u)|)^{\frac{N}{N-p}} \right)^{\frac{N-p}{N}}$$

where $a \in L^{N/p}(\Omega)$ by assumption and

$$\int_{\Omega} (|u_n|^{p-1} |T(u_n - u)|)^{\frac{N}{N-p}} = \int_{\Omega} |u_n|^{p^* - \frac{N}{N-p}} |T(u_n - u)|^{\frac{N}{N-p}} \quad (3.2)$$

$$\leq \left[\int_{\Omega} |u_n|^{(p^* - \frac{N}{N-p})q} \right]^{1/q} \left[\int_{\Omega} |T(u_n - u)|^{\frac{N}{N-p}q'} \right]^{1/q'} \quad (3.3)$$

$$= \left(\int_{\Omega} |u_n|^{p^*} \right)^{\frac{p-1}{p}} \left(\int_{\Omega} |T(u_n - u)|^{p^*} \right)^{1/p}, \quad (3.4)$$

where

$$q := \frac{p^*}{p^* - \frac{N}{N-p}} = \frac{(N-p)p^*}{(N-p)p^* - N} = \frac{Np}{Np - N} = \frac{p}{p-1} = p'.$$

By the same reasoning as before, we see that $\int_{\Omega} |a| |u_n|^{p-1} |T(u_n - u)| \rightarrow 0$ as $n \rightarrow \infty$ and therefore that $I_2 \rightarrow 0$. Putting all of this together, we find that

$$\lim_{n \rightarrow \infty} \int_{\Omega} (|\nabla u_n|^{p-2} \nabla u_n - |\nabla u_n|^{p-2} \nabla u) \cdot \nabla T(u_n - u) = 0.$$

Applying Theorem B with $\Omega_k = \Omega$, we may extract a subsequence (also denoted (u_n)) such that $\nabla u_n \rightarrow \nabla u$ almost everywhere on Ω . By this same theorem, we also have (i). Since (v_n) is bounded in $W^{1,p}(\Omega)$, the Sobolev inequality tells us that $\{|v_n|^p\}_{n \in \mathbb{N}}$ is bounded in $L^{\frac{N}{N-p}}(\Omega) \cong (L^{N/p}(\Omega))^*$. Recalling that $v_n \rightarrow 0$ pointwise almost everywhere

on Ω , an application of Theorem 2.1 shows that

$$\lim_{n \rightarrow \infty} \int_{\Omega} a(x) |v_n|^p = 0.$$

But then, (i) together with the Brézis-Lieb lemma implies

$$\begin{aligned} \phi_{\infty}(v_n) &= \int_{\Omega} \left(\frac{|\nabla v_n|^p}{p} - \mu \frac{|\nabla v_n|^{p^*}}{p^*} \right) \\ &= \frac{1}{p} \|u_n - u\|^p - \frac{\mu}{p^*} \|u_n - u\|_{L^{p^*}(\Omega)}^{p^*} \\ &= \frac{1}{p} (\|u_n\|^p - \|u\|^p) - \frac{\mu}{p^*} \left(\|u_n\|_{L^{p^*}(\Omega)}^{p^*} - \|u\|_{L^{p^*}(\Omega)}^{p^*} \right) + o(1) \\ &= \frac{1}{p} (\|u_n\|^p - \|u\|^p) - \frac{\mu}{p^*} \left(\|u_n\|_{L^{p^*}(\Omega)}^{p^*} - \|u\|_{L^{p^*}(\Omega)}^{p^*} \right) \\ &\quad + \int_{\Omega} a (|u_n|^p - |u|^p) + o(1) \\ &= \phi(u_n) - \phi(u) + o(1) \\ &= c - \phi(u) + o(1). \end{aligned}$$

This proves (ii). Now, we claim that $D\phi(u) = 0$. More precisely, we check that $D\phi(u)(h) = 0$ for all $h \in W_0^{1,p}(\Omega)$. Since $D\phi(u_n) \rightarrow 0$ strongly in $W^{-1,p'}(\Omega)$, it would actually suffice to show that

$$\lim_{n \rightarrow \infty} D\phi(u_n)(h) = D\phi(u)(h), \quad \forall h \in W_0^{1,p}(\Omega).$$

Let $h \in W_0^{1,p}(\Omega)$ be given and write

$$\begin{aligned} D\phi(u_n)(h) - D\phi(u)(h) &= \int_{\Omega} \left(|\nabla u_n|^{p-2} \nabla u_n \cdot \nabla h + a |u_n|^{p-2} u_n h - \mu |u_n|^{p^*-2} u_n h \right) \\ &\quad - \int_{\Omega} \left(|\nabla u|^{p-2} \nabla u \cdot \nabla h + a |u|^{p-2} u h - \mu |u|^{p^*-2} u h \right). \end{aligned}$$

As above, an application of Theorem 2.1 shows that

$$\lim_{n \rightarrow \infty} \int_{\Omega} |\nabla u_n|^{p-2} \nabla u_n \cdot \nabla h = \int_{\Omega} |\nabla u|^{p-2} \nabla u \cdot \nabla h,$$

and

$$\lim_{n \rightarrow \infty} \int_{\Omega} \mu |u_n|^{p^*-2} u_n h = \int_{\Omega} \mu |u|^{p^*-2} u h$$

Next, we claim that the family $\{|u_n|^{p-2} u_n h\}_{n \in \mathbb{N}}$ is bounded in $L^{\frac{N}{N-p}}(\Omega)$, where $\frac{N}{N-p}$ is the Hölder conjugate exponent of $\frac{N}{p}$. As in (3.2)-(3.4), we have

$$\begin{aligned} \int_{\Omega} (|u_n|^{p-1} |h|)^{\frac{N}{N-p}} &= \int_{\Omega} |u_n|^{p^* - \frac{N}{N-p}} |h|^{\frac{N}{N-p}} \\ &\leq \left[\int_{\Omega} |u_n|^{(p^* - \frac{N}{N-p})q} \right]^{1/q} \left[\int_{\Omega} |h|^{\frac{N}{N-p} q'} \right]^{1/q'} \\ &= \left(\int_{\Omega} |u_n|^{p^*} \right)^{\frac{p-1}{p}} \left(\int_{\Omega} |h|^{p^*} \right)^{1/p}. \end{aligned}$$

After a final application of Theorem 2.1, we infer that $D\phi(u_n)(h) \rightarrow D\phi(u)(h)$ for every fixed $h \in W_0^{1,p}(\Omega)$. Consequently, $D\phi(u) = 0$ in $W^{-1,p'}(\Omega)$. By similar convergence arguments, it readily follows that

$$D\phi_{\infty}(v_n) = D\phi(v_n) + o(1) = D\phi(u_n) - D\phi(u) + o(1) = o(1)$$

when considered in $W^{-1,p'}(\Omega)$. This completes the proof. $\square$

Finally, we cite a final result from Meruci-Willem which can be thought of as an analogue of the above for *rescalings* of the sequence (u_n) .

LEMMA 3.2. Assume $(\dagger)$ holds. Let (y_n) and (λ_n) be sequences in $\bar{\Omega}$ and $(0, \infty)$, respectively, such that

$$\frac{\text{dist}(y_n, \partial\Omega)}{\lambda_n^1} \xrightarrow{n \rightarrow \infty} \infty. \quad (3.5)$$

Suppose further that we have a sequence (u_n) in $\mathcal{D}^{1,p}(\mathbb{R}^N)$ such that the rescaled sequence

$$v_n(x) := \lambda_n^{\frac{N-p}{N}} u_n(\lambda_n x + y_n)$$

in $\mathcal{D}^{1,p}(\mathbb{R}^N)$ satisfies

- (1) $v_n \rightharpoonup v$ in $\mathcal{D}^{1,p}(\mathbb{R}^N)$ and pointwise almost everywhere on $\mathbb{R}^N$;
- (2) $\phi_{\infty}(u_n) \rightarrow c$ and $D\phi(u_n) \rightarrow 0$ in $W^{-1,p'}(\Omega)$ as $n \rightarrow \infty$.

Then $D\phi_{\infty}(v) = 0$. Furthermore, after passing to a subsequence, we have $\nabla v_n \rightarrow \nabla v$ almost everywhere on $\mathbb{R}^N$. Moreover,

- (i) $\lim_{n \rightarrow \infty} (\|u_n\|^p - \|w_n\|^p) = \|v\|^p$;
- (ii) $\phi_{\infty}(w_n) \rightarrow c - \phi_{\infty}(v)$ as $n \rightarrow \infty$; and
- (iii) $D\phi_{\infty}(w_n) \rightarrow 0$ in $W^{-1,p'}(\Omega)$ as $n \rightarrow \infty$

with

$$w_n(z) := u_n(z) - \lambda_n^{\frac{p-N}{N}} v \left(\frac{z - y_n}{\lambda_n} \right).$$

PROOF. We once again follow, with slight modification, the proof of Lemma 3.6 in Mercuri-Willem [3]. Note that (v_n) is bounded in $\mathcal{D}^{1,p}(\mathbb{R}^N)$ since it converges weakly. Let $\{B_k\}_{k \in \mathbb{N}}$ be an exhaustion of $\mathbb{R}^N$ by open balls, centered at the origin, of radius $k \geq 1$. For any such k , we claim that $D\phi_\infty(v_n) \rightarrow 0$ strongly in $W^{-1,p'}(B_k)$. Given a test function $h \in C_c^\infty(\mathbb{R}^N)$, we define

$$h_n(z) := \lambda_n^{\frac{p-N}{p}} h\left(\frac{z - y_n}{\lambda_n}\right).$$

Suppose now that $h \in C_c^\infty(B_k)$. By the assumption in (3.5), we have $h_n \in C_c^\infty(\Omega)$ for all n sufficiently large. Hence, a change of variables together with the Friedrich-Poincaré inequality gives

$$\begin{aligned} |D\phi_\infty(v_n)(h)| &= |D\phi_\infty(u_n)(h_n)| \leq \|D\phi_\infty(u_n)\|_{W^{-1,p'}(\Omega)} \|h_n\|_{W^{1,p}(\Omega)} \\ &\leq C \|D\phi_\infty(u_n)\|_{W^{-1,p'}(\Omega)} \|h_n\| \\ &= C \|D\phi_\infty(u_n)\|_{W^{-1,p'}(\Omega)} \|h\| \\ &\leq C \|D\phi_\infty(u_n)\|_{W^{-1,p'}(\Omega)} \|h\|_{W^{1,p}(B_k)}. \end{aligned}$$

It follows that $D\phi_\infty(v_n) \rightarrow 0$ in $W^{-1,p'}(B_k)$ as $n \rightarrow \infty$, for each $k \geq 1$. Let $T : \mathbb{R} \rightarrow \mathbb{R}$ be as in Theorem B and fix $k \geq 1$. Let $\rho \in C_c^\infty(\mathbb{R}^N)$ be such that $0 \leq \rho \leq 1$, $\rho \equiv 1$ on B_k , and $\rho \equiv 0$ outside B_{k+1} . For $n \in \mathbb{N}$, we define

$$f_n : \mathbb{R}^N \rightarrow \mathbb{R}^N, \quad f_n := |\nabla v_n|^{p-2} \nabla v_n - |\nabla v|^{p-2} \nabla v.$$

It is not hard to check that $f_n \cdot \nabla T(u_n - v) \geq 0$ on all of $\mathbb{R}^N$. Thus,

$$\int_{B_k} f_n \cdot \nabla T(v_n - v) = \int_{B_k} f_n \cdot \rho \nabla T(v_n - v) \leq \int_{\mathbb{R}^N} f_n \cdot \rho \nabla T(v_n - v).$$

Now, assuming that $\int_{\mathbb{R}^N} f_n \cdot \rho \nabla T(v_n - v) \rightarrow 0$ as $n \rightarrow \infty$, Theorem B would ensure the existence of a subsequence (v_n) with $\nabla v_n \rightarrow \nabla v$ almost everywhere on $\mathbb{R}^N$. In fact, we would also have (i) by this same theorem. A simple calculation verifies that

$$\int_{\mathbb{R}^N} f_n \cdot \rho \nabla T(v_n - v) = \underbrace{\int_{\mathbb{R}^N} f_n \cdot \nabla [T(v_n - v)\rho]}_{=: I_1} - \underbrace{\int_{\mathbb{R}^N} T(v_n - v) f_n \cdot \nabla \rho}_{=: I_2}.$$

As in the previous lemma, we are now reduced to verifying that $I_1, I_2 \rightarrow 0$ as $n \rightarrow \infty$. For I_2 , an application of Hölder's inequality gives the estimate

$$|I_2| \leq \int_{\mathbb{R}^N} |T(v_n - v) f_n \cdot \nabla \rho| \leq \|f_n\|_{L^{\frac{p}{p-1}}(\mathbb{R}^N; \mathbb{R}^N)} \|T(v_n) \nabla \rho\|_{L^p(\mathbb{R}^N; \mathbb{R}^N)}$$

where

$$\int_{\mathbb{R}^N} |T(v_n - v)|^p |\nabla \rho|^p \rightarrow 0 \quad \text{as } n \rightarrow \infty$$

by the dominated convergence theorem. Hence, $I_2 \rightarrow 0$. To see that $I_1 \rightarrow 0$ as well, we first note that

$$I_1 = D\phi_\infty(v_n)(\rho T(v_n - v)) + \underbrace{\mu \int_{\mathbb{R}^N} |v_n|^{p^*-2} v_n \rho T(v_n - v)}_{=: J_1} \quad (3.6)$$

$$- \underbrace{\int_{\mathbb{R}^N} |\nabla v|^{p-2} \nabla v \cdot \nabla [\rho T(v_n - v)]}_{=: J_2}. \quad (3.7)$$

Now, ρ has compact support and T is a bounded Lipschitz function with bounded derivative. Since $D\phi_\infty(v_n) \rightarrow 0$ strongly in $W^{-1,p'}(B_k)$ for all $k \geq 1$, it is not hard to see that

$$D\phi_\infty(v_n)(\rho T(v_n - v)) \rightarrow 0, \quad \text{as } n \rightarrow \infty.$$

To deal with J_1 , we first apply Hölder's inequality:

$$\begin{aligned} |J_1| &\leq \left\| |v_n|^{p^*-1} \right\|_{L^{\frac{p^*}{p^*-1}}(\mathbb{R}^N; \mathbb{R}^N)} \left\| \rho T(v_n - v) \right\|_{L^{p^*}(\mathbb{R}^N)} \\ &= \|v_n\|_{L^{p^*}(\mathbb{R}^N)}^{p^*-1} \left(\int_{\mathbb{R}^N} |\rho| |T(v_n - v)|^{p^*} \right)^{1/p^*} \end{aligned}$$

where

$$\lim_{n \rightarrow \infty} \int_{\mathbb{R}^N} |\rho| |T(v_n - v)|^{p^*} = 0$$

by the dominated convergence theorem. It follows that $J_1 \rightarrow 0$ as $n \rightarrow \infty$. Treating J_2 , we write

$$\begin{aligned} \int_{\mathbb{R}^N} |\nabla v|^{p-2} \nabla v \cdot \nabla [\rho T(v_n - v)] &= \int_{\mathbb{R}^N} |\nabla v|^{p-2} \rho \nabla v \cdot \nabla T(v_n - v) \\ &\quad + \int_{\mathbb{R}^N} |\nabla v|^{p-2} T(v_n - v) \nabla v \cdot \nabla \rho. \end{aligned}$$

As before, it is clear from Theorem 2.1 that

$$\lim_{n \rightarrow \infty} \int_{\mathbb{R}^N} |\nabla v|^{p-2} T(v_n - v) \nabla v \cdot \nabla \rho = 0.$$

On the other hand,

$$\int_{\mathbb{R}^N} |\nabla v|^{p-2} \rho \nabla v \cdot \nabla T(v_n - v) = \int_{\mathbb{R}^N} |\nabla v|^{p-2} \rho \nabla v \cdot T'(v_n - v) \nabla(v_n - v)$$

for each $n \in \mathbb{N}$. Since T' is bounded and $\nabla v_n \rightarrow \nabla v$ weakly in $L^p(\mathbb{R}^N; \mathbb{R}^N)$, we must also have $T'(v_n - v) \nabla(v_n - v) \rightarrow 0$ in $L^p(\mathbb{R}^N; \mathbb{R}^N)$. That is,

$$\lim_{n \rightarrow \infty} \int_{\mathbb{R}^N} |\nabla v|^{p-2} \rho \nabla v \cdot T'(v_n - v) \nabla(v_n - v) = 0$$

and so $J_2 \rightarrow 0$. By our earlier remarks, we may now assume that $\nabla v_n \rightarrow \nabla v$ almost everywhere on $\mathbb{R}^N$ and that (i) holds. Much like in Lemma 3.1 (and, of course, Mercuri-Willem [3]), the Brézis-Lieb lemma together with (i) implies that

$$\begin{aligned}\phi_\infty(w_n) &= \phi_\infty(v_n - v) = \phi_\infty(v_n) - \phi_\infty(v) + o(1) \\ &= \phi_\infty(u_n) - \phi_\infty(v) + o(1) \\ &= c - \phi_\infty(v) + o(1)\end{aligned}$$

whence we have (ii). Next, let us fix $h \in C_c^\infty(\mathbb{R}^N)$. Clearly, we have $D\phi_\infty(v_n)(h) \rightarrow 0$ as $n \rightarrow \infty$. On the other hand, we have

$$\begin{aligned}D\phi_\infty(v_n)(h) - D\phi_\infty(v)(h) &= \int_{\mathbb{R}^N} (|\nabla v_n|^{p-2} \nabla v_n \cdot \nabla h - \mu |v_n|^{p^*-2} v_n h) \\ &\quad - \int_{\mathbb{R}^N} (|\nabla v|^{p-2} \nabla v \cdot \nabla h - \mu |v|^{p^*-2} v h).\end{aligned}$$

Next, note that the $\{|\nabla v_n|^{p-1}\}_{n \geq 1}$ and $\{|v_n|^{p^*-1}\}_{n \geq 1}$ are bounded in the dual spaces of $L^p(\mathbb{R}^N; \mathbb{R}^N)$ and $L^{p^*}(\mathbb{R}^N)$, respectively. Since h has compact support, applying Theorem 2.1 in a now routine manner gives

$$\lim_{n \rightarrow \infty} \int_{\mathbb{R}^N} |\nabla v_n|^{p-2} \nabla v_n \cdot \nabla h = \int_{\mathbb{R}^N} |\nabla v|^{p-2} \nabla v \cdot \nabla h$$

and

$$\lim_{n \rightarrow \infty} \int_{\mathbb{R}^N} |v_n|^{p^*-2} v_n h = \int_{\mathbb{R}^N} |v|^{p^*-2} v h.$$

Thus, $D\phi_\infty(v) = 0$ in the dual of $\mathcal{D}^{1,p}(\mathbb{R}^N)$. Finally, convergence arguments that are by now familiar show that

$$D\phi_\infty(w_n)(g) \rightarrow 0 \quad \text{as } n \rightarrow \infty$$

uniformly in $g \in C_c^\infty(\Omega)$ having $\|g\| = 1$. This establishes (iii) and the proof is complete. $\square$

4. The Proof of Theorem 1.1

We are now ready to illustrate the method used by Mercuri-Willem [3] in the proof of Theorem 1.1. In the spirit of their proof, we shall divide the proof into several steps for the sake of readability. Recall that $\|\cdot\|$ by default denotes the $\mathcal{D}^{1,p}(\mathbb{R}^N)$ norm and *not* the $W_0^{1,p}(\Omega)$ -norm.

STEP 1. *There exists a subsequence, also denoted (u_n) , and $v_0 \in W_0^{1,p}(\Omega)$ such that $u_n \rightarrow v_0$ in weakly $W_0^{1,p}(\Omega)$ and strongly in $L^p(\Omega)$, u_n converges pointwise almost everywhere to*

v_0 on Ω , and $D\phi(v_0) = 0$. Furthermore, the sequence in $W_0^{1,p}(\Omega)$ given by

$$u_n^1 := u_n - v_0$$

is bounded in $W_0^{1,p}(\Omega)$ and satisfies

- (1) $\nabla u_n^1 \rightarrow 0$ almost everywhere on Ω ;
- (2) $\|u_n^1\|^p = \|u_n\|^p - \|v_0\|^p + o(1)$;
- (3) $\lim_{n \rightarrow \infty} \phi_\infty(u_n^1) = \lim_{n \rightarrow \infty} \phi(u_n) - \phi(v_0)$; and
- (4) $D\phi_\infty(u_n^1) \rightarrow 0$ in $W^{-1,p'}(\Omega)$ as $n \rightarrow \infty$.

PROOF OF STEP 1. Since the sequence (u_n) is bounded in $W_0^{1,p}(\Omega)$, we may obviously extract a subsequence converging weakly to some v_0 in $W_0^{1,p}(\Omega)$. Furthermore, since Ω is a smoothly bounded domain, the embedding

$$W_0^{1,p}(\Omega) \hookrightarrow L^p(\Omega)$$

is compact by the Rellich-Kondrachov theorem. Passing to yet another subsequence, we might as well assume that u_n converges *strongly* to some function w in $L^p(\Omega)$ and that $\lim_{n \rightarrow \infty} u_n(x) = w(x)$ for almost every $x \in \Omega$. In particular, u_n converges weakly to w in $L^p(\Omega)$. However, weak convergence in $W_0^{1,p}(\Omega)$ implies weak convergence in $L^p(\Omega)$. This means that $u_n \rightharpoonup v_0$ in $L^p(\Omega)$; by the uniqueness of weak limits we must then have $v_0 = w$ almost everywhere in Ω .

By virtue of Lemma 3.1, it is immediate that $D\phi(v_0) = 0$. Consequently, it remains only to verify the asymptotic identities in (2)-(4). This turns out to be quite easy since Lemma 3.1 directly implies both (3) and (4). As for (2), we have by this same theorem that

$$\lim_{n \rightarrow \infty} (\|u_n\|^p - \|u_n^1\|^p) = \|v_0\|^p.$$

It follows that

$$\begin{aligned} \|u_n^1\|^p + \|v_0\|^p - \|u_n\|^p &= \|v_0\|^p - (\|u_n\|^p - \|u_n^1\|^p) \\ &= \|v_0\|^p - \|v_0\|^p + o(1) = o(1). \end{aligned}$$

This completes the proof. We also point out that (u_n^1) is a sequence in $W_0^{1,p}(\Omega)$ which may be considered (using standard extension results) as a sequence in $W^{1,p}(\mathbb{R}^N)$. In fact, since the sequence $(u_n)_{n \in \mathbb{N}}$ is a Palais-Smale sequence, the auxiliary sequence $(u_n^1)_{n \in \mathbb{N}}$ will also be bounded in $W_0^{1,p}(\mathbb{R}^N)$. $\square$

STEP 2. Theorem 1.1 holds true in the case where $u_n^1 \rightarrow 0$ strongly in $L^p(\Omega)$.

PROOF OF STEP 2. In light of Step 1, we know that $D\phi_\infty(u_n^1)$ converges to 0 in $W^{-1,p'}(\Omega)$ as $n \rightarrow \infty$. In particular, we know that

$$\phi'_\infty(u_n^1)(h) = \int_{\mathbb{R}^N} \left(|\nabla u_n^1|^{p-2} \nabla u_n^1 \cdot \nabla h - \mu |u_n^1|^{p^*-2} u_n^1 h \right) \rightarrow 0, \quad \text{as } n \rightarrow \infty,$$

for every fixed $h \in W_0^{1,p}(\Omega)$. Note that this expression makes sense as $W_0^{1,p}(\Omega)$ can be interpreted as a subspace of $\mathcal{D}^{1,p}(\mathbb{R}^N)$ by the observation in Remark 1.2. Furthermore, for every integer $n \geq 1$ there holds

$$\begin{aligned} |D\phi_\infty(u_n^1)(u_n^1)| &\leq \sup_{n \in \mathbb{N}} \|u_n^1\| \|D\phi_\infty(u_n^1)\|_{W^{-1,p'}(\Omega)} \\ &\leq \sup_{n \in \mathbb{N}} \|u_n^1\|_{W^{1,p}(\Omega)} \|D\phi_\infty(u_n^1)\|_{W^{-1,p'}(\Omega)} \rightarrow 0, \end{aligned}$$

as $n \rightarrow \infty$. In this last step, we have used the boundedness of (u_n^1) in $W_0^{1,p}(\Omega)$. Returning to our expression for $D\phi_\infty(u_n^1)(h)$, we see that

$$D\phi_\infty(u_n^1)(u_n^1) = \int_{\mathbb{R}^N} \left(|\nabla u_n^1|^p - \mu |u_n^1|^{p^*} \right) \rightarrow 0, \quad \text{as } n \rightarrow \infty.$$

However, $u_n^1 \rightarrow 0$ in $L^{p^*}(\Omega)$ would imply that, as $n \rightarrow \infty$,

$$\int_{\mathbb{R}^N} \mu |u_n^1|^{p^*} \rightarrow 0.$$

From the last two equations, we infer that

$$\int_{\mathbb{R}^N} |\nabla u_n^1|^p \rightarrow 0, \quad \text{as } n \rightarrow \infty.$$

In other words, $\nabla u_n \rightarrow \nabla v_0$ in $L^p(\Omega)$. It follows that

$$\|u_n - v_0\| \rightarrow 0 \quad \text{and} \quad \|u_n\| \rightarrow \|v_0\|$$

as $n \rightarrow \infty$. Since $u_n \rightarrow v_0$ in $L^p(\Omega)$ by Step 1, we have that $u_n^1 \rightarrow 0$ in $L^p(\Omega)$. Combining these facts shows that $u_n^1 \rightarrow 0$ in $W_0^{1,p}(\Omega)$. That is,

$$\lim_{n \rightarrow \infty} u_n = v_0 \quad \text{strongly in } W_0^{1,p}(\Omega)$$

whence $\phi(v_0) = \lim_{n \rightarrow \infty} \phi(u_n)$. Furthermore, Step 1 states that

$$D\phi(v_0)(h) = 0, \quad \forall h \in W_0^{1,p}(\Omega).$$

In particular, this holds for all $h \in C_c^\infty(\Omega)$. This means precisely that v_0 solves problem (1.3) and Theorem 1.1 follows with $k = 0$. $\square$

Next, we will consider a family of Lévy concentration functions. For every index $n \geq 1$, we define a real valued function Q_n on $[0, \infty)$ via the formula

$$Q_n(r) := \sup_{y \in \bar{\Omega}} \int_{B(y,r)} |u_n^1|^{p^*}.$$

Note that each Q_n is well defined since every $u_n^1 \in L^{p^*}(\mathbb{R}^N)$ by the Sobolev embedding theorem. By the same logic, since the $\{u_n^1\}_{n \in \mathbb{N}}$ are uniformly bounded in $W_0^{1,p}(\mathbb{R}^N)$, the family

$$\mathcal{F}_n := \{Q_n(\cdot)\}_{n \in \mathbb{N}}$$

defines (by grace of Proposition 2.3) a collection of continuous functions $\{f_n\}_{n \in \mathbb{N}}$ on $[0, \infty)$. Now, in light of Step 2, we may assume that $u_n^1 \not\rightarrow 0$ strongly in $L^{p^*}(\Omega)$. After passing to a subsequence, this means that there exists $\delta > 0$ such that

$$\inf_{n \in \mathbb{N}} \int_{\Omega} |u_n^1|^{p^*} > \delta.$$

Without loss of generality, we may assume that

$$0 < \delta < \left(\frac{S_p}{2\mu}\right)^{N/p} \quad (4.1)$$

where $S_p > 0$ is the best constant for which the Gagliardo-Nirenberg-Sobolev inequality (GNS) holds:

$$S_p \|w\|_{L^{p^*}(\mathbb{R}^N)}^p \leq \|\nabla w\|_{L^p(\mathbb{R}^N; \mathbb{R}^N)}^p, \quad \forall w \in \mathcal{D}^{1,p}(\mathbb{R}^N),$$

Now, every Q_n is a continuous function on $[0, \infty)$ satisfying both

$$Q_n(0) = 0 \quad \text{and} \quad \lim_{r \rightarrow \infty} Q_n(r) > \delta.$$

By the intermediate value theorem, there exists $r_n \in [0, \infty)$ such that $Q_n(r_n) = \delta$. Letting R_n be the set of all such r_n , we put $\lambda_n^1 := \inf R_n \geq 0$. Fixing n , let $(r_l)_{l \in \mathbb{N}}$ be a sequence in R_n converging to λ_n^1 as $l \rightarrow \infty$. By continuity, it is clear that

$$Q_n(\lambda_n^1) = \lim_{l \rightarrow \infty} Q_n(r_l) = \delta.$$

Furthermore, since $Q_n(0) = 0$, we necessarily have $\lambda_n^1 > 0$. Proceeding inductively, we obtain a sequence (λ_n^1) in $(0, \infty)$ such that

$$Q_n(\lambda_n^1) = \sup_{y \in \bar{\Omega}} \int_{B(y, \lambda_n^1)} |u_n^1|^{p^*} = \delta,$$

for each $n \in \mathbb{N}$. Once again fixing $n \in \mathbb{N}$, we may find a sequence $(\xi_{m,n}^1)_{m \in \mathbb{N}}$ in $\bar{\Omega}$ such that

$$\delta = \lim_{m \rightarrow \infty} \int_{B(\xi_{m,n}^1, \lambda_n^1)} |u_n^1|^{p^*}.$$

Passing to a subsequence if necessary, we may assume that $\xi_{m,n}^1 \rightarrow y_n^1 \in \bar{\Omega}$ as we let $m \rightarrow \infty$. This then yields

$$\delta = \lim_{m \rightarrow \infty} \int_{B(\xi_{m,n}^1, \lambda_n^1)} |u_n^1|^{p^*} = \int_{B(y_n^1, \lambda_n^1)} |u_n^1|^{p^*}, \quad \forall n \in \mathbb{N}.$$

STEP 3. For each $n \in \mathbb{N}$ consider the domain

$$\Omega_n := \frac{1}{\lambda_n^1} (\Omega - y_n^1).$$

A simple calculation shows that $x \in \Omega_n$ if and only if

$$\lambda_n^1 x + y_n^1 \in \Omega.$$

Next, we define a sequence $(v_n^1)_{n \in \mathbb{N}}$ in $W_0^{1,p}(\Omega_n) \subseteq W^{1,p}(\mathbb{R}^N)$ by the rule

$$v_n^1(x) := \left(\lambda_n^1\right)^{\frac{N-p}{p}} u_n^1(\lambda_n^1 x + y_n^1).$$

The family $\{v_n^1\}_{n \in \mathbb{N}}$ belongs to the space $W^{1,p}(\mathbb{R}^N)$ and is uniformly bounded in $\mathcal{D}^{1,p}(\mathbb{R}^N)$. Moreover, one has

$$\delta = \sup_{y \in \bar{\Omega}_n} \int_{B(y,1)} |v_n^1|^{p^*} = \int_{B(0,1)} |v_n^1|^{p^*}. \quad (4.2)$$

PROOF OF STEP 3. Since every v_n^1 is a rescaling of a function in $W^{1,p}(\mathbb{R}^N)$, it is clear that $\{v_n^1\}_{n \in \mathbb{N}} \subset W^{1,p}(\mathbb{R}^N) \subseteq \mathcal{D}^{1,p}(\mathbb{R}^N)$. To see that the $\{v_n^1\}_{n \in \mathbb{N}}$ is uniformly bounded in $\mathcal{D}^{1,p}(\mathbb{R}^N)$, it suffices to make the following change of variables:

$$\begin{aligned} \|v_n^1\|^p &= \|\nabla v_n^1\|_{L^p(\mathbb{R}^N; \mathbb{R}^N)}^p = \int_{\mathbb{R}^N} \left| \lambda_n^1 \cdot \left(\lambda_n^1\right)^{\frac{N-p}{p}} \nabla u_n^1(\lambda_n^1 x + y_n^1) \right|^p dx \\ &= \int_{\mathbb{R}^N} \left| \left(\lambda_n^1\right)^{\frac{N}{p}} \nabla u_n^1(\lambda_n^1 x + y_n^1) \right|^p dx \\ &= \int_{\mathbb{R}^N} (\lambda_n^1)^N |\nabla u_n^1(\lambda_n^1 x + y_n^1)|^p dx \\ &= \int_{\mathbb{R}^N} |\nabla u_n^1|^p \\ &\leq \|u_n^1\|_{W^{1,p}(\mathbb{R}^N)}^p. \end{aligned}$$

Since the $\{u_n^1\}_{n \in \mathbb{N}}$ are bounded in $W^{1,p}(\mathbb{R}^N)$ by Step 1, the family $\{v_n^1\}_{n \in \mathbb{N}}$ is therefore bounded in $\mathcal{D}^{1,p}(\mathbb{R}^N)$. Next, we verify (4.2). Given $n \in \mathbb{N}$, we have by construction that

$$\begin{aligned} \int_{B(0,1)} |v_n^1|^{p^*} &= \int_{B(0,1)} \left| (\lambda_n^1)^{\frac{N-p}{p}} u_n^1(\lambda_n^1 x + y_n^1) \right|^{\frac{Np}{N-p}} dx \\ &= \int_{B(0,1)} (\lambda_n^1)^N |u_n^1(\lambda_n^1 x + y_n^1)|^{p^*} dx \\ &= \int_{B(y_n^1, \lambda_n^1)} |u_n^1|^{p^*} = \delta. \end{aligned}$$

Since $y \in \overline{\Omega_n}$ if and only if

$$y = \frac{z - y_n^1}{\lambda_n^1}$$

for a unique $z \in \bar{\Omega}$, a similar argument yields

$$\begin{aligned} \sup_{y \in \overline{\Omega_n}} \int_{B(y,1)} |v_n^1|^{p^*} &= \sup_{y \in \overline{\Omega_n}} \int_{B(y,1)} \left| (\lambda_n^1)^{\frac{N-p}{p}} u_n^1(\lambda_n^1 x + y_n^1) \right|^{\frac{Np}{N-p}} dx \\ &= \sup_{y \in \overline{\Omega_n}} \int_{B(y,1)} (\lambda_n^1)^N |u_n^1(\lambda_n^1 x + y_n^1)|^{p^*} dx \\ &= \sup_{z \in \bar{\Omega}} \int_{B(z, \lambda_n^1)} |u_n^1|^{p^*} = \delta. \end{aligned}$$

This directly implies (4.2) and the proof is complete. $\square$

REMARK 4.1. Since Ω is bounded, the sequence of radii (λ_n^1) must also be bounded. To be more explicit, we must have $0 < \lambda_n^1 \leq 2 \operatorname{diam}(\Omega)$ for each $n \geq 1$. Furthermore, the sequence of centers (y_n^1) lives in $\bar{\Omega}$ and must also be bounded. Thus, after passing to another subsequence, we may find y^1 in $\bar{\Omega}$ and $\lambda^1 \geq 0$ such that

$$\lim_{n \rightarrow \infty} y_n^1 = y^1 \quad \text{and} \quad \lim_{n \rightarrow \infty} \lambda_n^1 = \lambda^1.$$

For each $n \in \mathbb{N}$, let us now consider the linear functional $D\phi_\infty(u_n^1)(\cdot)$ on $\mathcal{D}^{1,p}(\mathbb{R}^N)$. Since $W_0^{1,p}(\Omega)$ is a closed subspace of $\mathcal{D}^{1,p}(\mathbb{R}^N)$, every $D\phi_\infty(u_n^1)(\cdot)$ restricts to an element of the dual $W^{-1,p'}(\Omega)$. By a duality result analogous to the Riesz Representation Theorem

for L^p , we can find $f_n^1, \dots, f_n^N \in L^{p'}(\Omega)$ such that⁵

$$D\phi_\infty(u_n^1)(h) = \sum_{i=1}^N \int_{\Omega} f_n^i \partial_i h$$

for all $h \in W_0^{1,p}(\Omega)$. Next, consider the functions given by

$$g_n^i = (\lambda_n^1)^{N-\frac{N}{p}} f_n^i(\lambda_n^1 x + y_n^1), \quad \forall i = 1, \dots, N.$$

STEP 4. *Adhering to the notation and terminology developed above, one has*

$$D\phi_\infty(v_n^1)(h) = \sum_{i=1}^N \int_{\Omega_n} g_n^i \partial_i h$$

for all $h \in W_0^{1,p}(\Omega_n)$.

PROOF OF STEP 4. First, we fix a function $h \in W_0^{1,p}(\Omega_n)$ and define

$$\tilde{h}(z) = h\left(\frac{z - y_n^1}{\lambda_n^1}\right).$$

Notice that $\tilde{h} \in W_0^{1,p}(\Omega)$. Furthermore, we have the following equalities;

$$\lambda_n^1 \partial_i \tilde{h}(z) = \partial_i h\left(\frac{z - y_n^1}{\lambda_n^1}\right) \quad \text{and} \quad \lambda_n^1 \nabla \tilde{h}(z) = \nabla h\left(\frac{z - y_n^1}{\lambda_n^1}\right).$$

Similarly, we see that

$$\nabla v_n^1(x) = (\lambda_n^1)^{(N-p)/p+1} \nabla u_n^1(\lambda_n^1 x + y_n^1) = (\lambda_n^1)^{N/p} \nabla u_n^1(\lambda_n^1 x + y_n^1).$$

⁵Given general domain Ω and a continuous linear functional φ on $W_0^{1,p}(\Omega)$, we can find $f_0, f_1, \dots, f_N$ belonging to $L^{p'}(\Omega)$ such that

$$\varphi(w) = \int_{\Omega} f_0 w + \sum_{i=1}^N \int_{\Omega} f_i \partial_i w$$

for all $w \in W_0^{1,p}(\Omega)$. However, when Ω is bounded, one can dispense with the f_0 -term in the above. That is, we have $\varphi(w) = \sum_{i=1}^N \int_{\Omega} f_i^p \partial_i w$ for suitable $f_1, \dots, f_N$ in $L^{p'}(\Omega)$. This is not too surprising since $w \mapsto \|\nabla w\|_{L^p(\Omega)}$ is an equivalent norm on $W_0^{1,p}(\Omega)$. We refer the reader to Adams-Fournier [1] and Brézis [2] for a precise statement and proof.

Fix now an integer $n \geq 1$ and let $h \in W_0^{1,p}(\Omega_n)$ be given. It is easily seen that

$$\begin{aligned}
\sum_{i=1}^N \int_{\Omega_n} g_n^i \partial_i h &= \sum_{i=1}^N \int_{\Omega_n} (\lambda_n^1)^{N-\frac{N}{p}} f_n^i(\lambda_n^1 x + y_n^1) \partial_i h(x) \, dx \\
&= (\lambda_n^1)^{N-\frac{N}{p}-N} \sum_{i=1}^N \int_{\Omega} f_n^i(z) \partial_i h \left(\frac{z - y_n^1}{\lambda_n^1} \right) \, dz \\
&= (\lambda_n^1)^{-\frac{N}{p}} \sum_{i=1}^N \int_{\Omega} f_n^i(z) \partial_i h \left(\frac{z - y_n^1}{\lambda_n^1} \right) \, dz \\
&= (\lambda_n^1)^{1-N/p} \sum_{i=1}^N \int_{\Omega} f_n^i(z) \partial_i \tilde{h}(z) \, dz \\
&= (\lambda_n^1)^{1-N/p} D\phi_{\infty}(u_n^1)(\tilde{h}).
\end{aligned}$$

On the other hand, $D\phi_{\infty}(v_n^1)(h)$ is given by

$$\int_{\mathbb{R}^N} \left[|\nabla v_n^1|^{p-2} \nabla v_n^1 \cdot \nabla h - \mu |v_n^1|^{p^*-2} v_n^1 h \right]$$

which, when expanded, becomes

$$\begin{aligned}
&\int_{\mathbb{R}^N} |(\lambda_n^1)^{N/p} \nabla u_n^1(\lambda_n^1 x + y_n^1)|^{p-2} (\lambda_n^1)^{N/p} \nabla u_n^1(\lambda_n^1 x + y_n^1) \cdot \nabla h(x) \, dx - \int_{\mathbb{R}^N} \mu |v_n^1|^{p^*-2} v_n^1 h \\
&= (\lambda_n^1)^{N(p-1)/p} \int_{\mathbb{R}^N} |\nabla u_n^1(\lambda_n^1 x + y_n^1)|^{p-2} \nabla u_n^1(\lambda_n^1 x + y_n^1) \cdot \nabla h(x) \, dx \\
&\quad - \int_{\mathbb{R}^N} \mu |(\lambda_n^1)^{(N-p)/p} u_n^1(\lambda_n^1 x + y_n^1)|^{p^*-2} (\lambda_n^1)^{(N-p)/p} u_n^1(\lambda_n^1 x + y_n^1) h(x) \, dx \\
&= (\lambda_n^1)^{N(p-1)/p-N} \int_{\mathbb{R}^N} |\nabla u_n^1(z)|^{p-2} \nabla u_n^1(z) \cdot \nabla h \left(\frac{z - y_n^1}{\lambda_n^1} \right) \, dz \\
&\quad - (\lambda_n^1)^{(N-p)(p^*-1)/p} \int_{\mathbb{R}^N} \mu |u_n^1(\lambda_n^1 x + y_n^1)|^{p^*-2} u_n^1(\lambda_n^1 x + y_n^1) h(x) \, dx \\
&= (\lambda_n^1)^{-N/p} \int_{\mathbb{R}^N} |\nabla u_n^1(z)|^{p-2} \nabla u_n^1(z) \cdot \nabla h \left(\frac{z - y_n^1}{\lambda_n^1} \right) \, dz \\
&\quad - (\lambda_n^1)^{(N-p)(p^*-1)/p-N} \int_{\mathbb{R}^N} \mu |u_n^1(z)|^{p^*-2} u_n^1(z) h \left(\frac{z - y_n^1}{\lambda_n^1} \right) \, dz \\
&= (\lambda_n^1)^{1-N/p} \int_{\mathbb{R}^N} |\nabla u_n^1(z)|^{p-2} \nabla u_n^1(z) \cdot \nabla \tilde{h}(z) \, dz - (\lambda_n^1)^{1-N/p} \int_{\mathbb{R}^N} \mu |u_n^1(z)|^{p^*-2} u_n^1(z) \tilde{h}(z) \, dz.
\end{aligned}$$

After some cleaning up, this last expression is equal to

$$(\lambda_n^1)^{1-N/p} \int_{\mathbb{R}^N} \left[|\nabla u_n^1(z)|^{p-2} \nabla u_n^1 \cdot \nabla \tilde{h} - \mu |u_n^1|^{p^*-2} u_n^1 \tilde{h} \right],$$

which is precisely

$$(\lambda_n^1)^{1-N/p} D\phi_\infty(u_n^1)(\tilde{h}).$$

Combining these identities, we find that for all $h \in W_0^{1,p}(\Omega_n)$:

$$D\phi_\infty(v_n^1)(h) = (\lambda_n^1)^{1-N/p} D\phi_\infty(u_n^1)(\tilde{h}) = \sum_{i=1}^N \int_{\Omega_n} g_n^i \partial_i h.$$

□

In light of the observations made in Remark 4.1, we may assume that $\Omega_n \rightarrow \Omega_\infty$, as $n \rightarrow \infty$, where $\Omega_\infty \subseteq \mathbb{R}^N$ is

- a rescaled translation of Ω if $\lambda^1 > 0$;
- all of $\mathbb{R}^N$ if $\lambda^1 = 0$ and $y^1 \in \Omega$; or
- a half space if $\lambda^1 = 0$ and $y^1 \in \partial\Omega$.

Here, the convergence should be understood in the sense of indicator functions, which is equivalent to the compact-open topology. However, a geometric picture of this convergence is enough for our purposes.

STEP 5. *The sequence (v_n^1) is uniformly bounded in $\mathcal{D}^{1,p}(\mathbb{R}^N)$ and, after passing to a subsequence, we may assume that $v_n^1 \rightarrow v_1$ for some $v_1 \in \mathcal{D}^{1,p}(\mathbb{R}^N)$ and that*

$$\lim_{n \rightarrow \infty} v_n^1(x) = v_1(x)$$

for almost every $x \in \mathbb{R}^N$.

PROOF OF STEP 5. We already know from Step 3 that (v_n^1) is bounded in $\mathcal{D}^{1,p}(\mathbb{R}^N)$. The remaining assertions follow at once from Corollary 2.5 and Corollary 2.7. □

STEP 6. *The weak $\mathcal{D}^{1,p}(\mathbb{R}^N)$ -limit v_1 of v_n^1 is non-zero.*

PROOF OF STEP 6. We argue by contradiction. If $v_1 = 0$ in $\mathcal{D}^{1,p}(\mathbb{R}^N)$, then v_1 vanishes almost everywhere in $\mathbb{R}^N$. Recall that the $\{v_n^1\}_{n \in \mathbb{N}}$ are uniformly bounded in $\mathcal{D}^{1,p}(\mathbb{R}^N)$. Since the family $\{v_n^1\}_{n \in \mathbb{N}}$ is bounded in $\mathcal{D}^{1,p}(\mathbb{R}^N)$, the Gagliardo-Nirenberg-Sobolev inequality (GNS) ensures that the $\{v_n^1\}_{n \in \mathbb{N}}$ also form a bounded family of functions in $L^{p^*}(\mathbb{R}^N)$.

Now, given an open ball $B \subset \mathbb{R}^N$, we see that $\{v_n^1\}_{n \in \mathbb{N}}$ is a uniformly bounded sequence in $W^{1,p}(B)$. By the Rellich-Kondrachov theorem, we may assume that $v_n^1 \rightarrow w$ strongly in $L^p(B)$. Then, passing to a subsequence yet again, we must have $v_n^1 \rightarrow w$ pointwise almost everywhere in this ball B . Hence, $w = v_1$ almost everywhere on B . In short, we have found a subsequence (v_n^1) that converges to 0 strongly in $L^p(B)$. By taking an exhaustion of $\mathbb{R}^N$ by open balls of radius $m \geq 1$ and using a diagonal argument, we

find a subsequence (also denoted by (v_n^1)) that converges to 0 in $L^p(B_m)$, for each $m \in \mathbb{N}$. Clearly, this means that $v_n^1 \rightarrow 0$ in $L_{\text{loc}}^p(\mathbb{R}^N)$ as $n \rightarrow \infty$.

By using Step 1 together with a duality result from Brézis [2] (Proposition 9.20 therein), it follows that

$$\|D\phi_\infty(u_n^1)\|_{W^{-1,p'}(\Omega)} = \sum_{i=1}^N \int_{\Omega} |f_n^i|^{p'} \quad (4.3)$$

$$= \sum_{i=1}^N \int_{\Omega_n} |g_n^i|^{p'} \rightarrow 0, \quad \text{as } n \rightarrow \infty. \quad (4.4)$$

Next, recall that Ω_∞ denotes the “limiting domain” of the Ω_n . By definition, if $y \in \Omega_\infty$, then y belongs to Ω_n for all sufficiently large n . Fix now a point $y \in \Omega_\infty$ and consider the ball $B(y, 1)$, which may not be contained in Ω_∞ . For a given test function $h \in C_c^\infty(\mathbb{R}^N)$ with $\text{supp}(h) \subset B(y, 1)$, Hölder’s inequality with exponent $\frac{N}{p}$, gives

$$\begin{aligned} \int_{\mathbb{R}^N} |h|^p |v_n^1|^{p^*} &= \int_{\mathbb{R}^N} |v_n^1|^{p^*-p} \cdot |h v_n^1|^p \\ &\leq \left(\int_{\text{supp}(h)} |v_n^1|^{p^*} \right)^{p/N} \left(\int_{\mathbb{R}^N} |h v_n^1|^{p^*} \right)^{\frac{N-p}{N}}. \end{aligned}$$

From this, an application of Sobolev’s inequality shows that

$$\int_{\mathbb{R}^N} |h|^p |v_n^1|^{p^*} \leq \left(\int_{\text{supp}(h)} |v_n^1|^{p^*} \right)^{p/N} \left(\int_{\mathbb{R}^N} |h v_n^1|^{p^*} \right)^{\frac{N-p}{N}} \quad (4.5)$$

$$\leq S_p^{-1} \left(\int_{\text{supp}(h)} |v_n^1|^{p^*} \right)^{p/N} \left(\int_{\mathbb{R}^N} |\nabla(h v_n^1)|^p \right). \quad (4.6)$$

Because v_n^1 converges *strongly* to 0 in $L_{\text{loc}}^p(\mathbb{R}^N)$, one then has

$$\int_{\Omega_n} |\nabla(h v_n^1)|^p = \int_{\mathbb{R}^N} |h|^p |\nabla v_n^1|^p + o(1) \quad (4.7)$$

$$= \int_{\mathbb{R}^N} |\nabla v_n^1|^{p-2} \nabla v_n^1 \cdot \nabla (|h|^p v_n^1) + o(1) \quad (4.8)$$

$$= \langle \phi'_\infty(v_n^1), |h|^p v_n^1 \rangle + \mu \int_{\mathbb{R}^N} |v_n^1|^{p^*-2} (v_n^1)^2 |h|^p + o(1) \quad (4.9)$$

$$= \sum_{i=1}^N \int_{\Omega_n} g_n^i \partial_i (|h|^p v_n^1) + \mu \int_{\mathbb{R}^N} |h|^p |v_n^1|^{p^*} + o(1) \quad (4.10)$$

$$\leq \mu \int_{B(y,1)} |h|^p |v_n^1|^{p^*} + o(1). \quad (4.11)$$

In the above series of equation, equality (4.7) follows from Hölder's inequality. Equation (4.9) follows (4.3)-(4.4) with an application of Hölder's inequality after noting that $\partial_i(|h|^p v_n^1)$ is bounded in $L^p(\mathbb{R}^N)$.⁶ From (4.2) and (4.5)-(4.6), it follows that

$$\begin{aligned} \int_{\Omega_n} |h \nabla v_n^1|^p &\leq \mu S_p^{-1} \delta^{\frac{p}{N}} \int_{\mathbb{R}^N} |\nabla(h v_n^1)|^p + o(1) \\ &\leq \frac{1}{2} \int_{\mathbb{R}^N} |\nabla(h v_n^1)|^p + o(1) \end{aligned}$$

where the last inequality follows from property (4.1). Whence we conclude that

$$\lim_{n \rightarrow \infty} \int_{\Omega_n} |h \nabla v_n^1|^p = 0.$$

This means that $\nabla v_n^1 \rightarrow 0$ in $L_{\text{loc}}^p(B)$, for any ball B of radius 1 with center in Ω_∞ . After covering Ω_∞ with countably many balls, each of radius 1 with center in Ω_∞ , we infer that ∇v_n^1 converges to 0 in $L_{\text{loc}}^p(\Omega_\infty)$, as $n \rightarrow \infty$. Similarly, $\|\nabla v_n^1\|_{L^p(B)} \rightarrow 0$ on any ball B compactly contained in the complement of Ω_∞ , as $n \rightarrow \infty$. To summarize, we have shown that $\nabla v_n^1 \rightarrow 0$ in $L_{\text{loc}}^p(\mathbb{R}^N)$ whence $v_n^1 \rightarrow 0$ in $W_{\text{loc}}^{1,p}(\mathbb{R}^N)$. Finally, the Sobolev inequality implies that

$$\lim_{n \rightarrow \infty} v_n^1 = 0 \quad \text{in } L_{\text{loc}}^{p^*}(\mathbb{R}^N),$$

which directly contradicts (4.2). $\square$

STEP 7. *The radii converge to 0 as $n \rightarrow \infty$, i.e. $\lambda^1 = \lim_{n \rightarrow \infty} \lambda_n^1 = 0$.*

PROOF OF STEP 7. Arguing by contradiction, let us assume that $\lambda^1 > 0$. Since $v_n^1 \rightarrow v_1$ weakly in $\mathcal{D}^{1,p}(\mathbb{R}^N)$, Proposition 2.4 states that

$$\lim_{n \rightarrow \infty} \int_{\mathbb{R}^N} \nabla v_n^1 \cdot \varphi = \int_{\mathbb{R}^N} \nabla v_1 \cdot \varphi, \quad \forall \varphi \in L^{p'}(\mathbb{R}^N; \mathbb{R}^N). \quad (4.12)$$

⁶We notice that $\partial_i(|h|^p v_n^1) = v_n^1 \partial_i(|h|^p) + |h|^p \partial_i v_n^1$. To see that $v_n^1 \partial_i(|h|^p)$ is bounded in $L^p(\mathbb{R}^N)$, we recall that v_n^1 is bounded in $L^{p^*}(\mathbb{R}^N)$ and $h \in C_c^\infty(\mathbb{R}^N)$. Similarly, $|h|^p \partial_i v_n^1$ is bounded in $L^p(\mathbb{R}^N)$ since $\partial_i v_n^1$ is bounded in $L^p(\mathbb{R}^N)$ and $h \in C_c^\infty(\mathbb{R}^N)$.

For the moment, let us assume that $\varphi \in C_c^\infty(\mathbb{R}^N; \mathbb{R}^N)$. Then, a direct computation shows that

$$\begin{aligned} \int_{\mathbb{R}^N} \nabla v_n^1 \cdot \varphi &= \int_{\mathbb{R}^N} (\lambda_n^1)^{\frac{N-p}{p}} \cdot \lambda_n^1 \nabla u_n^1(\lambda_n^1 x + y_n^1) \cdot \varphi(x) \, dx \\ &= \int_{\mathbb{R}^N} (\lambda_n^1)^N \nabla u_n^1(\lambda_n^1 x + y_n^1) \cdot \varphi(x) \, dx \\ &= \int_{\mathbb{R}^N} \nabla u_n^1(z) \cdot \varphi\left(\frac{z - y_n^1}{\lambda_n^1}\right) \, dz. \end{aligned}$$

Now, Step 1 ensures that $\nabla u_n^1 \rightarrow 0$ pointwise almost everywhere on Ω . Actually, since $\{u_n^1\}_{n \in \mathbb{N}} \subset W_0^{1,p}(\Omega)$, we know that $\nabla u_n^1 \rightarrow 0$ pointwise almost everywhere on $\mathbb{R}^N$. Moreover, it is not hard to see that the sequence (∇u_n^1) is uniformly bounded in $L^p(\mathbb{R}^N; \mathbb{R}^N)$.⁷ Since $\lambda_n^1 \rightarrow \lambda^1 > 0$ and $y_n^1 \rightarrow y^1 \in \bar{\Omega}$, there exists a compact set $\Lambda \subset \mathbb{R}^N$ such that

$$\text{supp}\left(\varphi\left(\frac{\cdot - y_n^1}{\lambda_n^1}\right)\right) \subseteq \Lambda, \quad \forall n \geq 1.$$

Using now that φ is bounded, an Application of Theorem 2.1 gives

$$\lim_{n \rightarrow \infty} \int_{\mathbb{R}^N} \nabla v_n^1 \cdot \varphi = \lim_{n \rightarrow \infty} \int_{\mathbb{R}^N} \nabla u_n^1(z) \cdot \varphi\left(\frac{z - y_n^1}{\lambda_n^1}\right) \, dz = 0.$$

Indeed, this is because for $M := \sup_{\mathbb{R}^N} |\varphi|$ one has

$$\begin{aligned} \left| \int_{\mathbb{R}^N} \nabla u_n^1(z) \cdot \varphi\left(\frac{z - y_n^1}{\lambda_n^1}\right) \, dz \right| &\leq \int_{\mathbb{R}^N} \left| \nabla u_n^1(z) \cdot \varphi\left(\frac{z - y_n^1}{\lambda_n^1}\right) \right| \, dz \\ &\leq \int_{\mathbb{R}^N} |\nabla u_n^1| M \mathbf{1}_\Lambda. \end{aligned}$$

Now, we relax the assumption that $\varphi \in C_c^\infty(\mathbb{R}^N; \mathbb{R}^N)$. Fix $g \in L^{p'}(\mathbb{R}^N; \mathbb{R}^N)$ and let $\varepsilon > 0$ be given. By density, there exists $\varphi \in C_c^\infty(\mathbb{R}^N; \mathbb{R}^N)$ such that

$$\|g - \varphi\|_{L^{p'}(\mathbb{R}^N; \mathbb{R}^N)} < \varepsilon.$$

Clearly, one has

$$\begin{aligned} \left| \int_{\mathbb{R}^N} \nabla v_n^1 \cdot g \right| &\leq \left| \int_{\mathbb{R}^N} \nabla v_n^1 \cdot \varphi \right| + \left| \int_{\mathbb{R}^N} \nabla v_n^1 \cdot (g - \varphi) \right| \\ &\leq \left| \int_{\mathbb{R}^N} \nabla v_n^1 \cdot \varphi \right| + \|v_n^1\| \|g - \varphi\|_{L^{p'}(\mathbb{R}^N; \mathbb{R}^N)}. \end{aligned}$$

⁷This is simply because weakly convergent sequences in $W_0^{1,p}(\Omega)$ are bounded.

Since $\{v_n^1\}_{n \in \mathbb{N}}$ is a bounded family in $\mathcal{D}^{1,p}(\mathbb{R}^N)$ and $\int_{\mathbb{R}^N} \nabla v_n^1 \cdot \varphi \rightarrow 0$ as $n \rightarrow \infty$, we infer that

$$\left| \int_{\mathbb{R}^N} \nabla v_n^1 \cdot g \right| < \varepsilon + \sup_{n \in \mathbb{N}} \|v_n^1\| \varepsilon$$

for all n large. Or, rather, that

$$\lim_{n \rightarrow \infty} \int_{\mathbb{R}^N} \nabla v_n^1 \cdot \varphi = 0, \quad \forall \varphi \in L^{p'}(\mathbb{R}^N; \mathbb{R}^N).$$

But by Proposition 2.4 this means that $v_n^1 \rightharpoonup 0$ in $\mathcal{D}^{1,p}(\mathbb{R}^N)$, which directly contradicts Step 6. It follows that $\lambda_n^1 \rightarrow 0$ as $n \rightarrow \infty$. Furthermore, the sequence in $\mathcal{D}^{1,p}(\mathbb{R}^N)$ given by

$$u_n^2(x) := u_n^1(x) - \left(\lambda_n^1\right)^{\frac{p-N}{N}} v_1 \left(\frac{x - y_n^1}{\lambda_n^1} \right)$$

satisfies

- (1) $\|u_n^2\|^p = \|u_n\|^p - \|v_0\|^p - \|v_1\|^p + o(1)$;
- (2) $\phi_\infty(u_n^2) \rightarrow \lim_{n \rightarrow \infty} \phi(u_n) - \phi(v_0) - \phi_\infty(v_1)$; and
- (3) $D\phi_\infty(u_n^2) \rightarrow 0$ in $W^{-1,p'}(\Omega)$.

□

Since $\lambda_n^1 \rightarrow 0$ as $n \rightarrow \infty$, we can rule out the case where Ω_∞ is a rescaled translation of Ω . This leaves us with only two possibilities:

- $y^1 \in \Omega$ and $\Omega_\infty = \mathbb{R}^N$; or
- $y_1 \in \partial\Omega$ and Ω_∞ is a half-space.

Let us now assume that we are in the first case, i.e. $\Omega_\infty = \mathbb{R}^N$; a similar argument (together with an analogue of Lemma 3.2 and a slightly different definition of ϕ_∞) applies in the second case. After applying Step 1 together with Lemma 3.2, we see that $D\phi_\infty(v_1) = 0$. That is, v_1 solves equation (1.4).

STEP 8. *If $w \in \mathcal{D}^{1,p}(\mathbb{R}^N)$ is a non-trivial critical point of ϕ_∞ , one has*

$$\phi_\infty(w) \geq \frac{\mu}{N} \left(\frac{S_p}{\mu} \right)^{N/p} > 0. \quad (4.13)$$

PROOF OF STEP 8. Let now $w \in \mathcal{D}^{1,p}(\mathbb{R}^N)$ be a non-trivial critical point of ϕ_∞ . By (GNS), this forces

$$0 = D\phi_\infty(w)(w) = \int_{\mathbb{R}^N} (|\nabla w|^p - \mu |w|^{p^*})$$

whence we have

$$S_p \|w\|_{L^{p^*}}^p \leq \|\nabla w\|_{L^p(\mathbb{R}^N; \mathbb{R}^N)} = \|w\| = \mu \|w\|_{L^{p^*}(\mathbb{R}^N)}^{p^*}.$$

Especially, this gives

$$\frac{S_p}{\mu} \leq \|w\|_{L^{p^*}(\mathbb{R}^N)}^{p^*-p} \text{ so that } \|w\|_{L^{p^*}(\mathbb{R}^N)} \geq \left(\frac{S_p}{\mu}\right)^{\frac{1}{p^*-p}}.$$

From this, we obtain

$$\begin{aligned} \phi_\infty(w) &= \frac{\|w\|^p}{p} - \frac{\mu \|w\|_{L^{p^*}}^{p^*}}{p^*} = \mu \|w\|_{L^{p^*}}^{p^*} \left(\frac{1}{p} - \frac{1}{p^*} \right) \\ &\geq \mu \left[\left(\frac{S_p}{\mu} \right)^{\frac{1}{p^*-p}} \right]^{p^*} \left(\frac{1}{N} \right) \\ &= \frac{\mu}{N} \left(\frac{S_p}{\mu} \right)^{N/p} > 0. \end{aligned}$$

In light of this, the iterative procedure can only construct finitely many sequences (v_n^i) , (λ_n^i) , and (y_n^i) . Afterwards, we would find ourselves in Step 2 thereby terminating the argument. We note that this part of the argument is standard and can be found explicitly in Struwe [5]-[6], Willem [7]-[8], and Mercuri-Willem [3]. $\square$

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